

Material Science Engineering

By AMIT MEENA

Introduction

- Materials are probably more deep seated in our culture than most of us realize.
- Transportation, housing, clothing, communication, recreation, and food production— virtually every segment of our everyday lives is influenced to one degree or another by materials.
- The development of many technologies that make our existence so comfortable has been intimately associated with the accessibility of suitable materials.

WHY STUDY MATERIALS SCIENCE

- Many an applied scientist or engineer, whether mechanical, civil, chemical, or electrical, is at one time or another exposed to a design problem involving materials, such as a transmission gear, the superstructure for a building, an oil refinery component, or an integrated circuit chip. Of course, materials scientists and engineers are specialists who are totally involved in the investigation and design of materials.
- Many times, a materials problem is one of selecting the right material from the thousands available. The final decision is normally based on several criteria. First, the in-service conditions must be characterized, for these dictate the properties required of the material. On only rare occasions does a material possess the maximum or ideal combination of properties. Thus, it may be necessary to trade one characteristic for another.
- The classic example involves strength and ductility; normally, a material having a high strength has only a limited ductility. In such cases, a reasonable compromise between two or more properties may be necessary.
- A second selection consideration is any deterioration of material properties that may occur during service operation. For example, significant reductions in mechanical strength may result from exposure to elevated temperatures or corrosive environments.
- Finally, probably the overriding consideration is that of economics: What will the finished product cost? A material may be found that has the ideal set of properties but is prohibitively expensive. Here again, some compromise is inevitable. The cost of a finished piece also includes any expense incurred during fabrication to produce the desired shape.

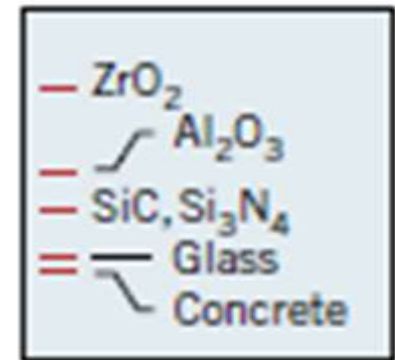
CLASSIFICATION OF MATERIALS

- Solid materials have been conveniently grouped into three basic categories: metals, ceramics, and polymers, a scheme based primarily on chemical makeup and atomic structure.
- Another category is advanced materials—those used in high-technology applications, such as semiconductors, biomaterials, smart materials, and nanoengineered materials;

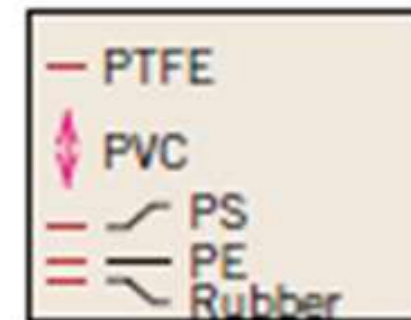
Metals



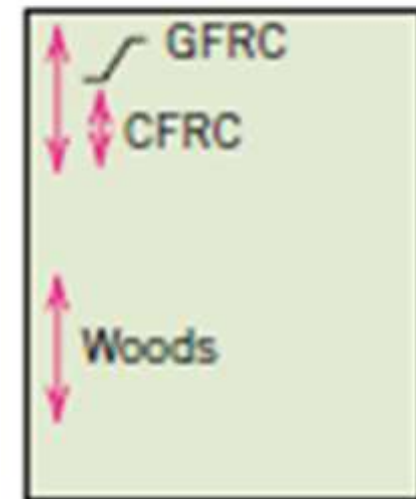
Ceramics



Polymers



Composites



ADVANCED MATERIALS

- Materials utilized in high-technology (or high-tech) applications are sometimes termed *advanced materials*.
- These advanced materials are typically traditional materials whose properties have been enhanced and also newly developed, high-performance materials. Furthermore, they may be of all material types (e.g., metals, ceramics, polymers) and are normally expensive.

ADVANCED MATERIALS :Smart Materials

- **Semiconductors**

- *Semiconductors* have electrical properties that are intermediate between those of electrical conductors (i.e., metals and metal alloys) and insulators (i.e., ceramics and polymers). Furthermore, the electrical characteristics of these materials are extremely sensitive to the presence of minute concentrations of impurity atoms, for which the concentrations may be controlled over very small spatial regions. Semiconductors have made possible the advent of integrated circuitry that has totally revolutionized the electronics and computer industries (not to mention our lives) over the past three decades.

- **Biomaterials**

- *Biomaterials* are employed in components implanted into the human body to replace diseased or damaged body parts. These materials must not produce toxic substances and must be compatible with body tissues (i.e., must not cause adverse biological reactions). All of the preceding materials—metals, ceramics, polymers, composites, and semiconductors— may be used as biomaterials.

ADVANCED MATERIALS : Smart Materials

- *Smart* (or *intelligent*) *materials* are a group of new and state-of-the-art materials now being developed that will have a significant influence on many of our technologies. The adjective *smart* implies that these materials are able to sense changes in their environment and then respond to these changes in predetermined manners—traits that are also found in living organisms. In addition, this *smart* concept is being extended to rather sophisticated systems that consist of both smart and traditional materials.
- Components of a smart material (or system) include some type of sensor (which detects an input signal) and an actuator (which performs a responsive and adaptive function). Actuators may be called upon to change shape, position, natural frequency, or mechanical characteristics in response to changes in temperature, electric fields, and/or magnetic fields.
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- Four types of materials are commonly used for actuators: shape-memory alloys, piezoelectric ceramics, magnetostrictive materials, and electrorheological/magnetorheological fluids.
- *Shape-memory alloys* are metals that, after having been deformed, revert to their original shape when temperature is changed
- *Piezoelectric ceramics* expand and contract in response to an applied electric field (or voltage); conversely, they also generate an electric field when their dimensions are altered .
- The behavior of *magnetostrictive materials* is analogous to that of the piezoelectrics, except that they are responsive to magnetic fields.
- Also, *electrorheological* and *magnetorheological fluids* are liquids that experience dramatic changes in viscosity upon the application of electric and magnetic fields, respectively.

Nanomaterials

- One new material class that has fascinating properties and tremendous technological promise is the *nanomaterials*, which may be any one of the four basic types—metals, ceramics, polymers, or composites. However, unlike these other materials, they are not distinguished on the basis of their chemistry but rather their size; the *nano* prefix denotes that the dimensions of these structural entities are on the order of a nanometer (10^{-9} m)—as a rule, less than 100 nanometers (nm; equivalent to the diameter of approximately 500 atoms).
- Prior to the advent of nanomaterials, the general procedure scientists used to understand the chemistry and physics of materials was to begin by studying large and complex structures and then investigate the fundamental building blocks of these structures that are smaller and simpler. This approach is sometimes termed *top-down science*.

- However, with the development of scanning probe microscopes , which permit observation of individual atoms and molecules, it has become possible to design and build new structures from their atomic-level constituents, one atom or molecule at a time (i.e., “materials by design”). This ability to arrange atoms carefully provides opportunities to develop mechanical, electrical, magnetic, and other properties that are not otherwise possible. We call this the *bottom-up approach*, and the study of the properties of these materials is termed *nanotechnology*.
- Some of the physical and chemical characteristics exhibited by matter may experience dramatic changes as particle size approaches atomic dimensions. For example, materials that are opaque in the macroscopic domain may become transparent on the nanoscale; some solids become liquids, chemically stable materials become combustible, and electrical insulators become conductors. Furthermore, properties may depend on size in this nanoscale domain. Some of these effects are quantum mechanical in origin, whereas others are related to *surface phenomena*—the proportion of atoms located on surface sites of a particle increases dramatically as its size decreases. Because of these unique and unusual properties, nanomaterials are finding niches in electronic, biomedical, sporting, energy production, and other industrial applications.

Unit 1

Crystalline Nature of Solids

INTRODUCTION

- Some of the important properties of solid materials depend on geometric atomic arrangements and also the interactions that exist among constituent atoms or molecules.
- The present discussion is devoted to the structure of materials, specifically, to some of the arrangements that may be assumed by atoms in the solid state.
- Within this framework, concepts of crystallinity and non-crystallinity are introduced. For crystalline solids, the notion of crystal structure is presented, specified in terms of a unit cell.
- The three common crystal structures found in metals are then detailed, along with the scheme by which crystallographic points, directions, and planes are expressed.

FUNDAMENTAL CONCEPTS

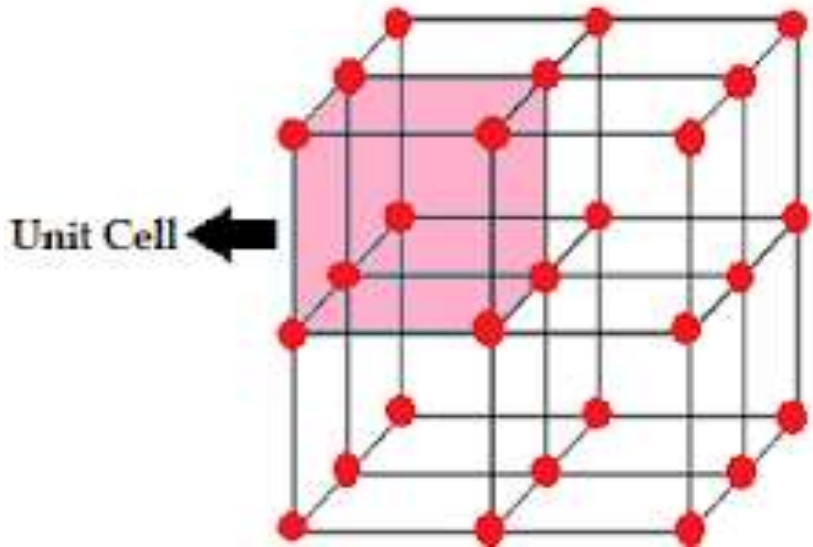
- A **crystalline** material is one in which the atoms are situated in a repeating or periodic array over large atomic distances—that is, long-range order exists, such that upon solidification, the atoms will position themselves in a repetitive three-dimensional pattern, in which each atom is bonded to its nearest neighbor atoms.
- All metals, many ceramic materials, and certain polymers form crystalline structures under normal solidification conditions.
- For those that do not crystallize, this long-range atomic order is absent; these *noncrystalline* or *amorphous* materials are discussed briefly at the end.

CRYSTAL STRUCTURE

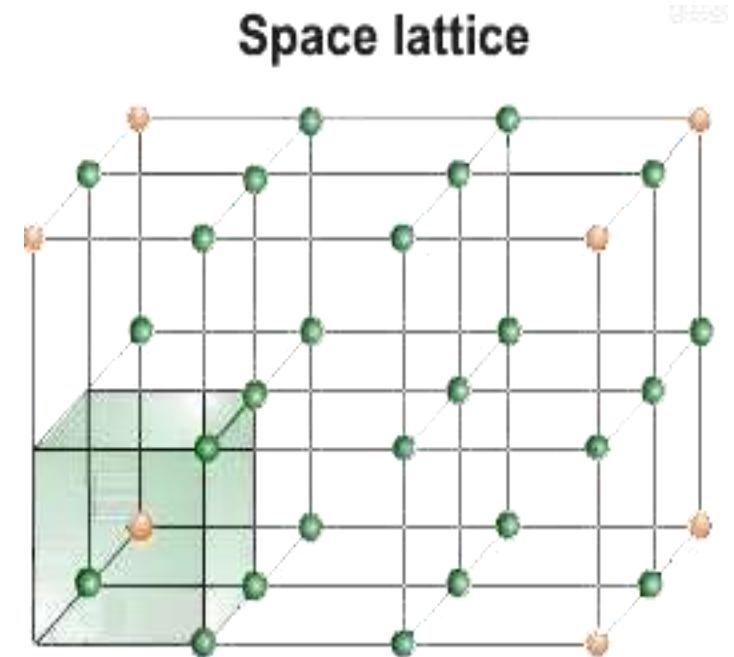
- Materials can be broadly classified as crystalline and non-crystalline solids. In a **crystal**, the arrangement of atoms is in a periodically repeating pattern whereas no such regularity of arrangement is found in a non-crystalline material.
- A crystalline solid can either be a single crystal, where the entire solid consists of only one crystal or an aggregate of many crystals separated by well-defined boundaries.
- In the latter form, the solid is said to be poly-crystalline.

Unit Cell and Space Lattice

- The metallic crystals can be considered as consisting of tiny blocks which are repeated in three dimensional pattern. The tiny block formed by the arrangement of a small group of atoms is called the **unit cell**. It is that volume of solid from which the entire crystal can be constructed by translational repetition in three dimensions.

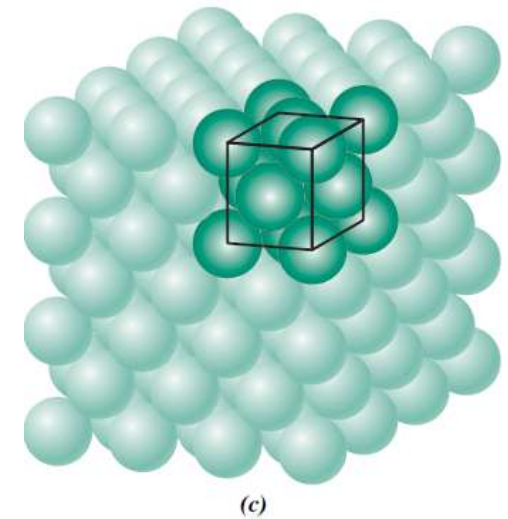
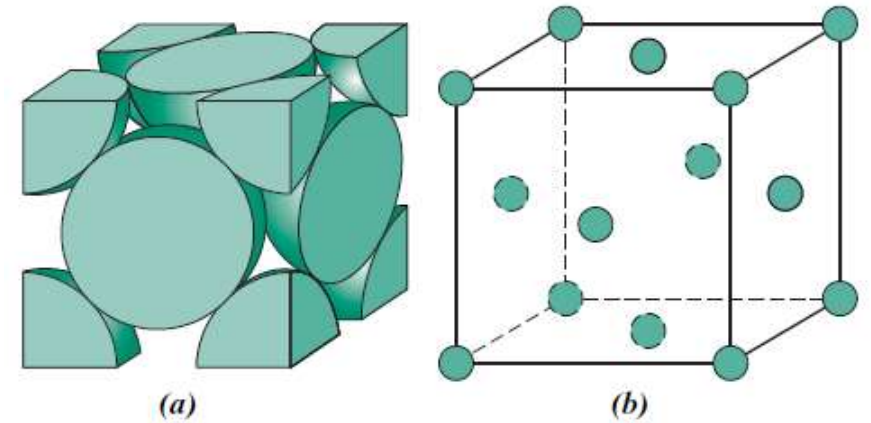


If each atom in a lattice is replaced by a point, then each point is called a lattice point and the arrangement of the points is referred to as the (three dimensional) lattice array. Thus a space lattice is defined as an array of points in three dimensions in which every point has surroundings identical to that every other point in the array. The distance between the atoms-points is called inter atomic or lattice spacing. A space lattice is a conventional geometrical basis by which crystal structures can be described. Every point of space lattice has identical surroundings.



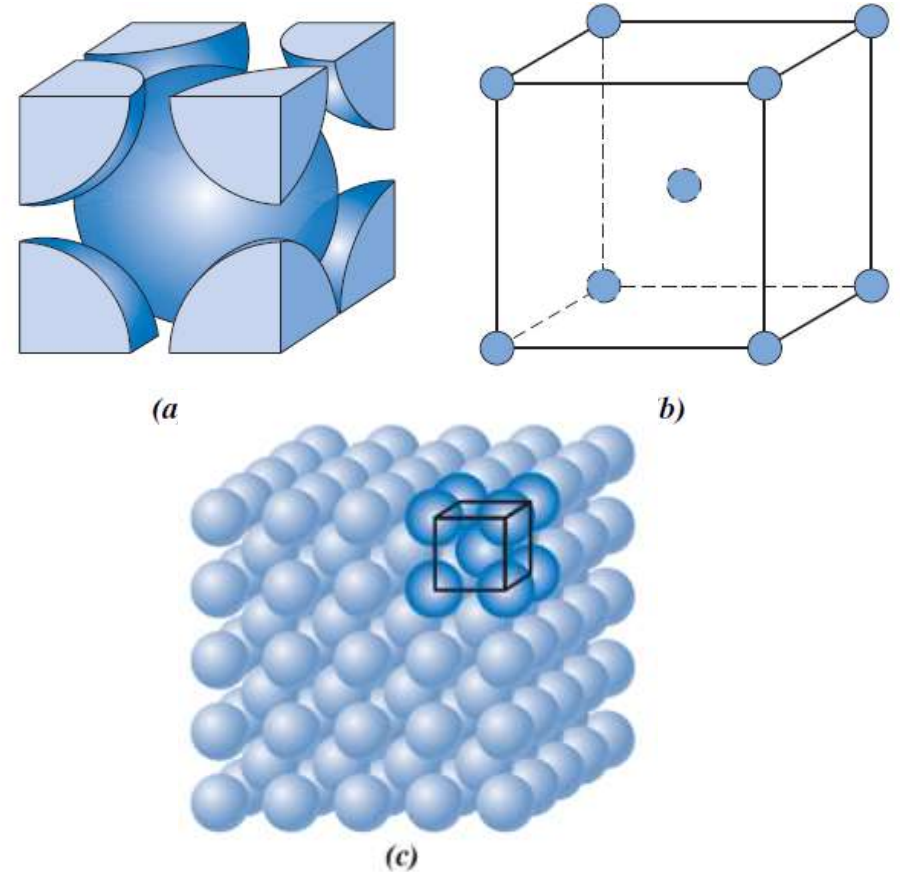
The Face-Centered Cubic Crystal Structure(FCC)

- The crystal structure found for many metals has a unit cell of cubic geometry, with atoms located at each of the corners and the centers of all the cube faces.
- It is aptly called the **face-centered cubic (FCC)** crystal structure.
- Some of the familiar metals having this crystal structure are copper, aluminum, silver, and gold .
- Figure *a* shows a hard-sphere model for the FCC unit cell, whereas in Figure *b* the atom centers are represented by small circles to provide a better perspective on atom positions.
- The aggregate of atoms in Figure *c* represents a section of crystal consisting of many
- FCC unit cells. These spheres or ion cores touch one another across a face diagonal;



The Body-Centered Cubic Crystal Structure(BCC)

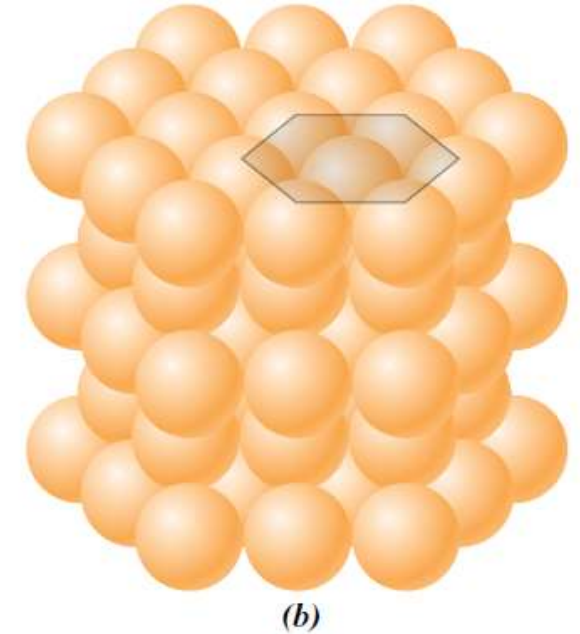
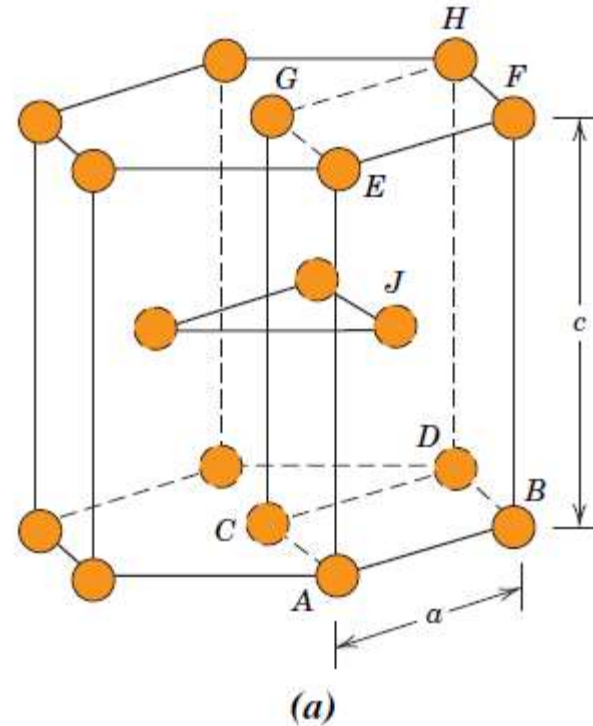
- Another common metallic crystal structure also has a cubic unit cell with atoms located at all eight corners and a single atom at the cube center.
- This is called a body-centered cubic (BCC) crystal structure. A collection of spheres depicting this crystal structure is shown in Figure c, whereas Figures a and b are diagrams of BCC unit cells with the atoms represented by hard-sphere and reduced-sphere models, respectively.
- Chromium, iron, tungsten, and several other metals exhibit a BCC structure.
- Each BCC unit cell has eight corner atoms and a single center atom, which is wholly contained within its cell



For the body-centered cubic crystal structure, (a) a hard-sphere unit cell representation, (b) a reduced-sphere unit cell, and (c) an aggregate of many atoms. [Figure (c) from W. G. Moffatt, G. W. Pearsall, and J.

The Hexagonal Close-Packed Crystal Structure(HCP)

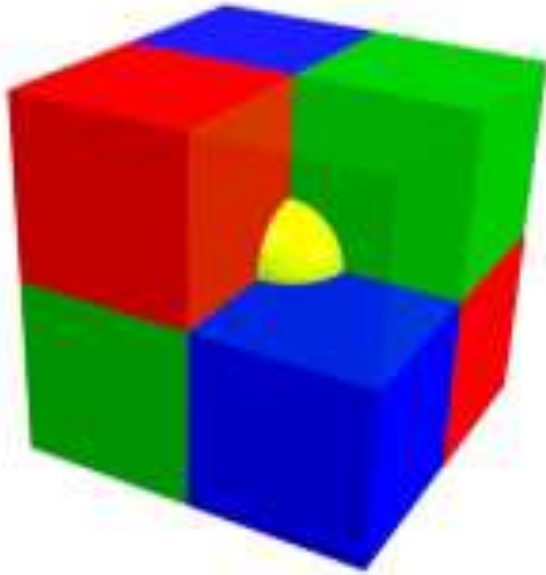
- Not all metals have unit cells with cubic symmetry; the final common metallic crystal
- structure to be discussed has a unit cell that is hexagonal.
- Figure *a* shows a reduced sphere unit cell for this structure, which is termed **hexagonal close-packed (HCP)**;
- An assemblage of several HCP unit cells is presented in Figure *b*. The top and bottom faces of the unit cell consist of six atoms that form regular hexagons and surround a single atom in the center.
- Another plane that provides three additional atoms to the unit cell is situated between the top and bottom planes. The atoms in this midplane have as nearest neighbors atoms in both of the adjacent two planes.



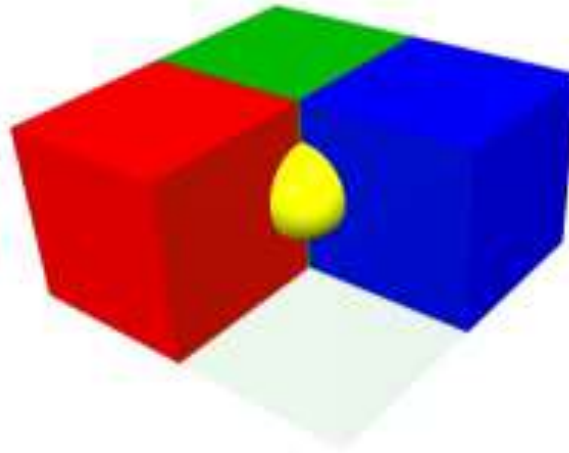
For the hexagonal close-packed crystal structure, (a) a reduced-sphere unit cell (a and c represent the short and long edge lengths, respectively), and (b) an aggregate of many atoms.

Contributions per Unit Cell

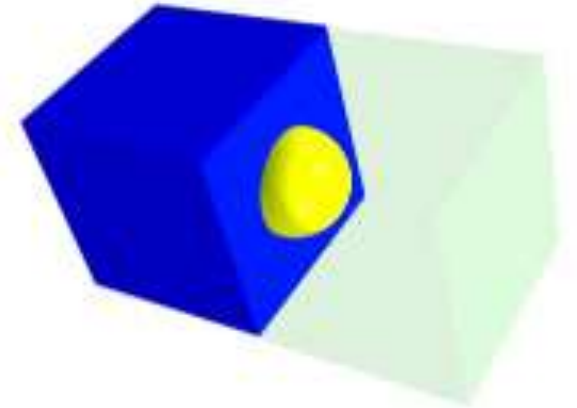
- The particles on the eight corners combine to contribute 1 particle to a unit cell. Particles in the six face centers combine to contribute 3 particles to a unit cell. Those on twelve edges combine to contribute 3 particles per unit cell.



A particle on a **corner** is shared by eight unit cells, so it contributes $1/8$ particle to each.



A particle on an edge is shared by four unit cells, so it contributes $1/4$ particle to each.



A particle on a face center is shared by two unit cells, so it contributes $1/2$ particle to each.

- Atoms that are entirely within a unit cell each contribute a full atom to the cell stoichiometry, but the atoms in the faces, corners, and edges are shared by more than one unit cell, and the number of atoms they contribute to each unit cell is given below

- $$\frac{8 \text{ corners}}{\text{unit cell}} \times \frac{1 \text{ atom}}{8 \text{ Corners}} = \frac{1 \text{ atom}}{\text{unit Cell}}$$

- Atoms located on the *corners* of the unit cell contribute 1 atom per unit cell.

- $$\frac{6 \text{ faces}}{\text{unit cell}} \times \frac{1 \text{ atom}}{2 \text{ Faces}} = \frac{3 \text{ atoms}}{\text{unit Cell}}$$

- Atoms located on the *faces* of the unit cell contribute 3 atoms per unit cell.

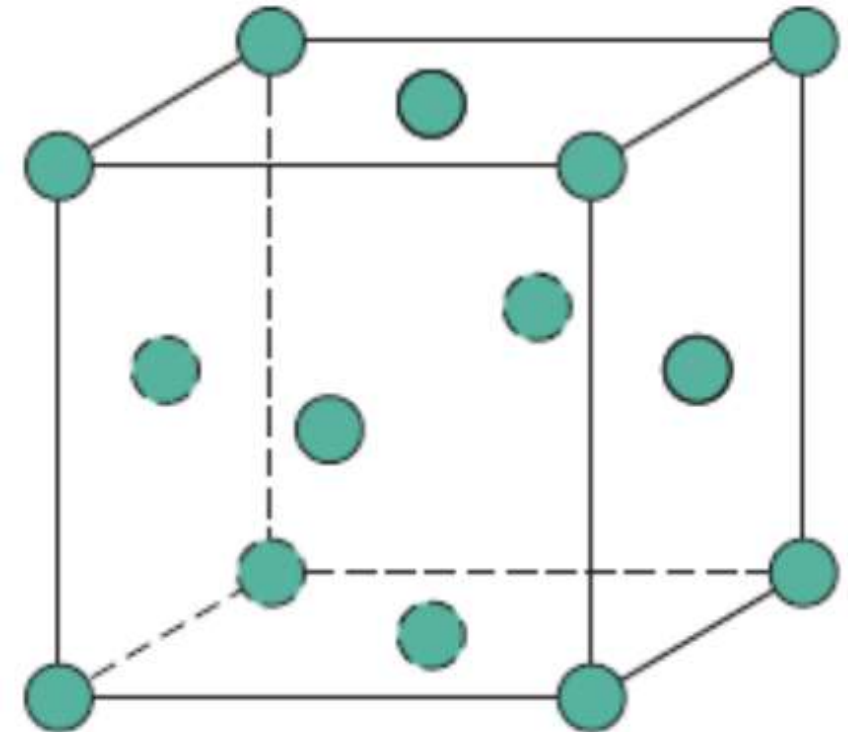
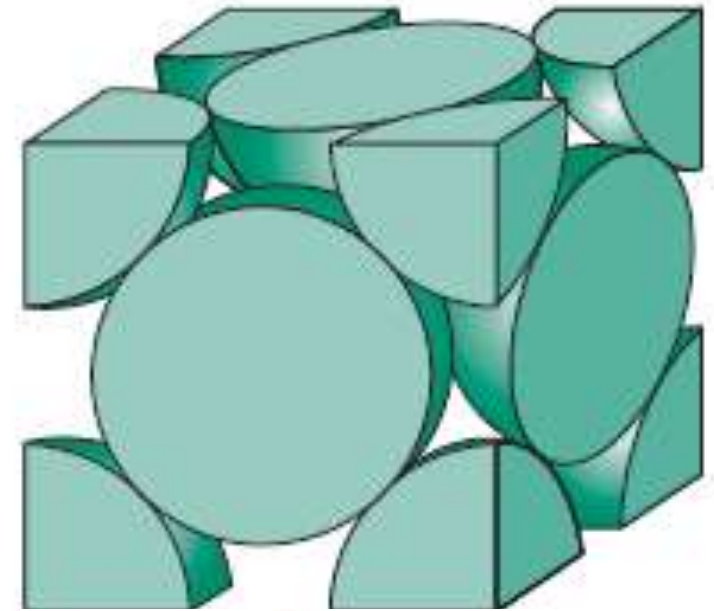
- $$\frac{12 \text{ edges}}{\text{unit cell}} \times \frac{1 \text{ atom}}{4 \text{ edges}} = \frac{3 \text{ atoms}}{\text{unit Cell}}$$

- Atoms located on the *edges* of the unit cell contribute 3 atoms per unit cell.

No of atoms in Unit Cell

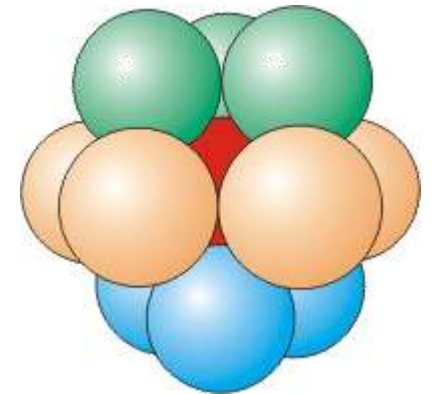
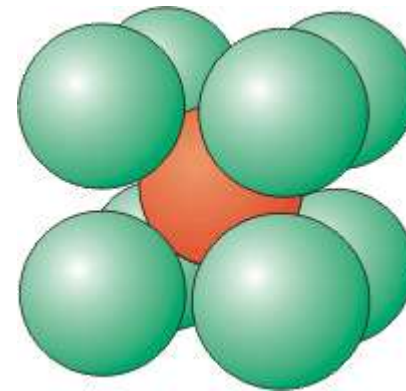
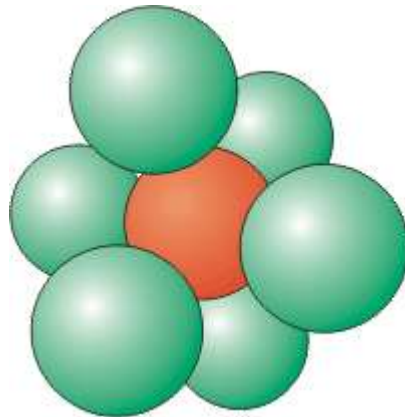
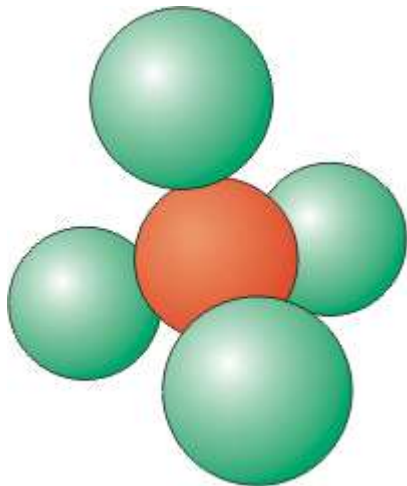
- we need to determine the number of atoms associated with each unit cell. Depending on an atom's location, it may be considered to be shared with adjacent unit cells—that is, only some fraction of the atom is assigned to a specific cell.
- For example, for cubic unit cells, an atom completely within the interior “belongs” to that unit cell, one at a cell face is shared with one other cell, and an atom residing at a corner is shared among eight.
- The number of atoms per unit cell, N , can be computed using the following formula:
- $$N = N_i + N_f/2 + N_c/8$$
- where
 - N_i the number of interior atoms
 - N_f the number of face atoms
 - N_c the number of corner atoms

- For the FCC crystal structure, there are eight corner atoms ($N_c = 8$), six face atoms ($N_f = 6$), and no interior atoms ($N_i = 0$). Thus, from Equation ,
- $N = 0 + 6/2 + 8/8 = 4$
- or a total of four whole atoms may be assigned to a given unit cell. This is depicted in Figure , where only sphere portions are represented within the confines of the cube.
- The cell is composed of the volume of the cube that is generated from the centers of the corner atoms, as shown in the figure.
- Corner and face positions are really equivalent—that is, translation of the cube corner from an original corner atom to the center of a face atom will not alter the cell structure.

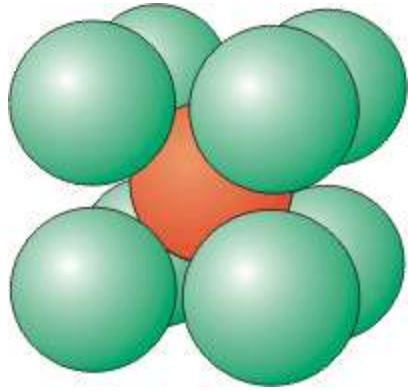
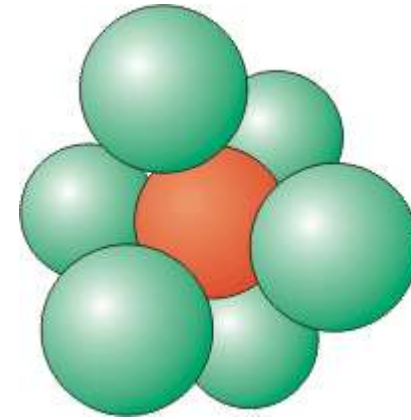


COORDINATION NUMBER

- The ***coordination number*** (CN) is the number of nearest neighbors of a given particle in the crystal lattice.
- It determines the nature of the bonding in a crystal.
- The most common coordination numbers are 4, 6, 8, and 12, respectively as shown below

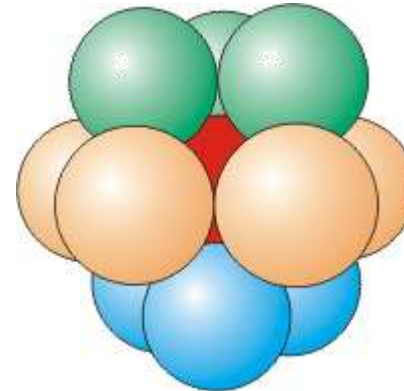


An atom on the corner of a simple cubic unit cell has a coordination number of six.



The atom in the center of a body-centered unit cell has a coordination number of eight.

Each atom in a face-centered cubic unit cell has a coordination number of 12. The 12 particles, which make up the *coordination sphere* of the red particle lie in three planes.

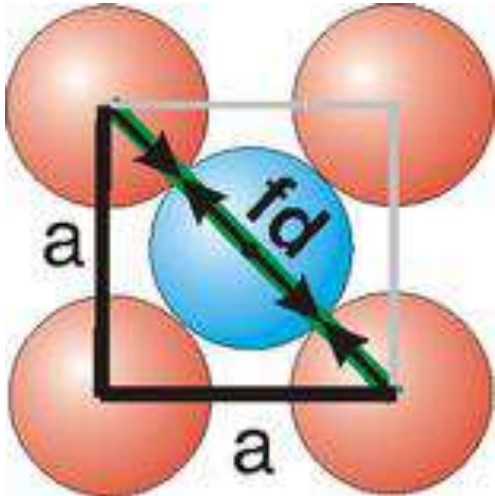
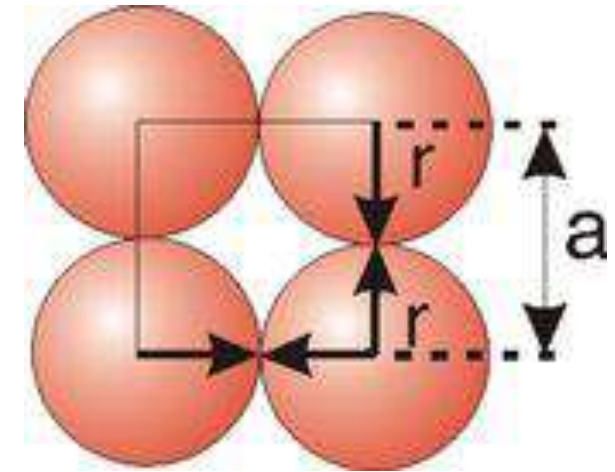


Atomic Radii

- Atoms are not hard spheres with distinct boundaries, so their sizes are not determined directly. However, the positions of the atoms in a solid can be determined by x-ray diffraction, and the sizes of the atoms are inferred from those distances.
- In this method, the radius of an atom is determined from the unit cell edge length (a), which is determined from the location of the atoms, and the assumption that the atoms touch as shown in Figures (next Slide).
- The radius obtained from the structure of a metallic solid is referred to as either the ***metallic radius*** or the ***atomic radius***

Atomic Radii

$2r = a$ for simple cubic structure
Here r = atomic radius ; a = cube length

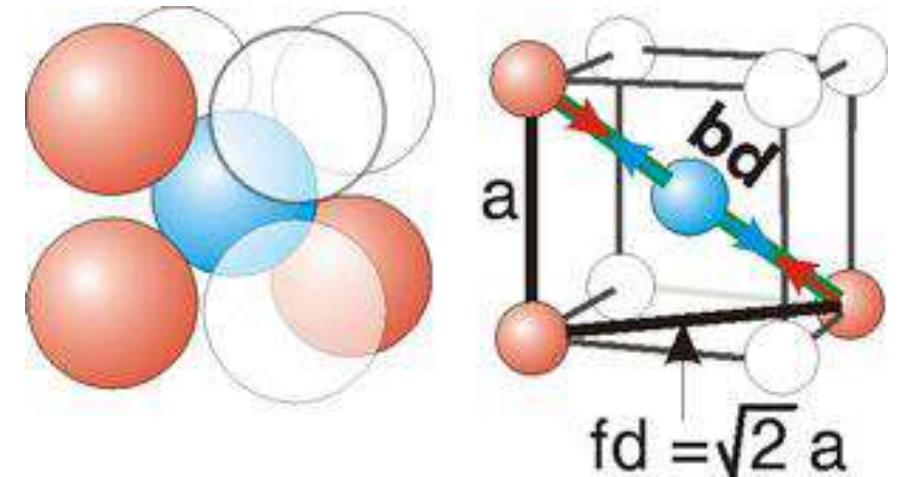


$$fd = 4r \rightarrow (fd)^2 = (a)^2 + (a)^2 \rightarrow (4r)^2 = 2(a)^2 \rightarrow r = \sqrt{2}a/4$$

For FCC structure

$$d = \sqrt{2}a \quad (bd)^2 = (d)^2 + (a)^2 \quad (4r)^2 = 3(a)^2$$

$r = \sqrt{3}a/4$ For BCC structure

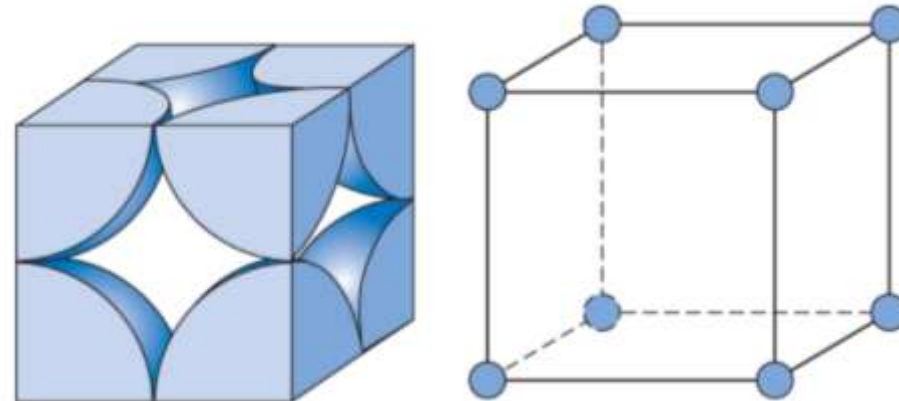


Determination of FCC Unit Cell Volume

- ***Solution***
- In the FCC unit cell illustrated, the atoms touch one another across a face-diagonal, the length of which is $4r$. Because the unit cell is a cube, its volume is a^3 , where a is the cell edge length.
- From the right triangle on the face,
 - $a^2 + a^2 = 4r^2$
 - or, solving for a ,
 - $a = 2r \sqrt{2}$
- The FCC unit cell volume VC may be computed from
 - $VC = a^3 = (2r \sqrt{2})^3 = 16 r^3 \sqrt{2}$
 - Similarly find out for SC BCC & HCP

Atomic packing factor (APF).

- The APF is the sum of the sphere volumes of all atoms within a unit cell (assuming the atomic hard-sphere model) divided by the unit cell volume
- $APF = \frac{\text{volume of atoms in a unit cell}}{\text{total unit cell volume}}$
- Simple Cubic Structure:-In this structure, atoms touch each other along the lattice, as shown in Fig. $a = 2r$ or $r = a/2$

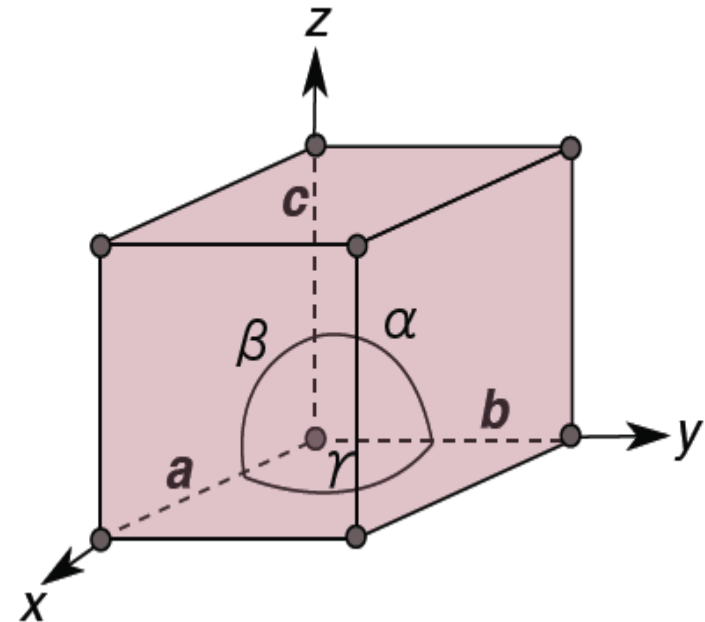


Computation of the APF for FCC

- $APF = \frac{\text{volume of atoms in a unit cell}}{\text{total unit cell volume}} = \frac{V_s}{V_c}$
- Both the total atom and unit cell volumes may be calculated in terms of the atomic radius r .
- The volume for a sphere $= \frac{4}{3}\pi r^3$
- and because there are four atoms per FCC unit cell, the total
- FCC atom (or sphere) volume is $V_s = (4) \times \frac{4}{3}\pi r^3 = \frac{16}{3}\pi r^3$
- From Example earlier, the total unit cell volume is $V_c = 16 r^3 \sqrt{2}$
- Therefore, the atomic packing factor is
- $APF = \frac{V_s}{V_c} = \frac{\frac{16}{3}\pi r^3}{16 r^3 \sqrt{2}} = 0.74$

Lattice geometry

- To define the geometry of the unit cell in 3 dimensions we choose a right-handed set of crystallographic axes, x , y , and z , which point along the edges of the unit cell. The origin of our coordinate system is at one of the lattice points.
- Lattice parameters
- The length of the unit cell along the x , y , and z direction are defined as a , b , and c . Alternatively, we can think of the sides of the unit cell in terms of vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$. The angles between the crystallographic axes are defined by:
 - α = the angle between $\mathbf{b}$ and $\mathbf{c}$
 - β = the angle between $\mathbf{a}$ and $\mathbf{c}$
 - γ = the angle between $\mathbf{a}$ and $\mathbf{b}$
- a , b , c , α , β , γ are collectively known as the **lattice parameters** (often also called 'unit cell parameters', or just 'cell parameters').



Crystal systems

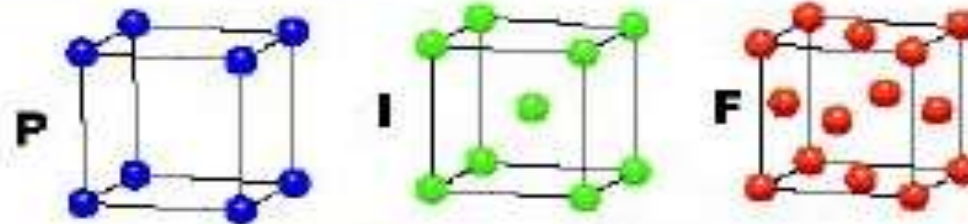
- The rotational symmetry of a crystal places constraints on the shape of the conventional unit cell we choose to describe the structure. On this basis we divide all structures into one of 7 crystal systems. For example, for crystals with 4 fold symmetry it will always be possible to choose a unit cell that has a square base with $a = b$ and $\gamma = 90^\circ$:
- There are 14 unique combinations of the 7 crystal systems with the possible types of primitive and non-primitive lattices. These are referred to as the 14 Bravais lattices.

Seven Crystal Structures

CUBIC

$$a = b = c$$

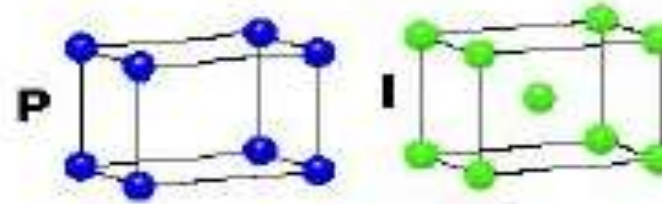
$$\alpha = \beta = \gamma = 90^\circ$$



TETRAGONAL

$$a = b \neq c$$

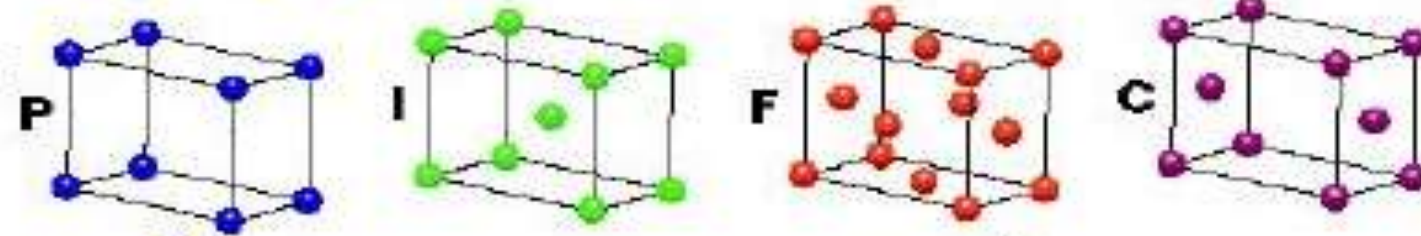
$$\alpha = \beta = \gamma = 90^\circ$$



ORTHORHOMBIC

$$a \neq b \neq c$$

$$\alpha = \beta = \gamma = 90^\circ$$

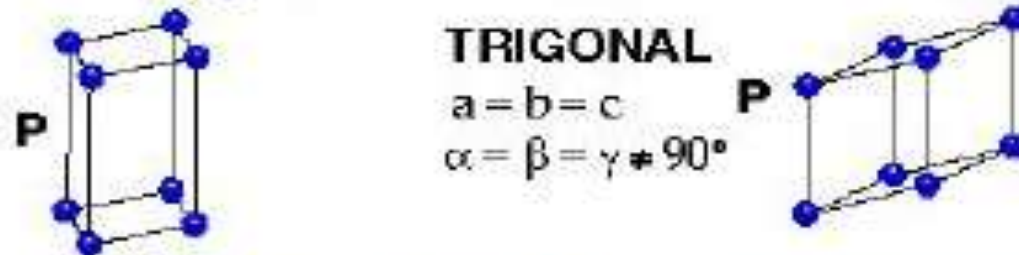


HEXAGONAL

$$a = b \neq c$$

$$\alpha = \beta = 90^\circ$$

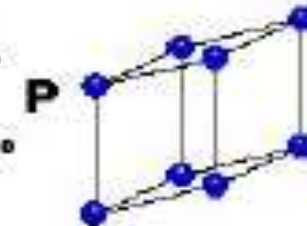
$$\gamma = 120^\circ$$



TRIGONAL

$$a = b = c$$

$$\alpha = \beta = \gamma \neq 90^\circ$$

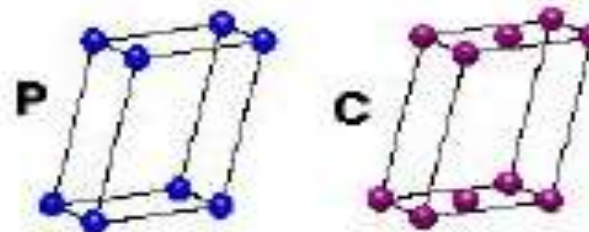


MONOCLINIC

$$a \neq b \neq c$$

$$\alpha = \gamma = 90^\circ$$

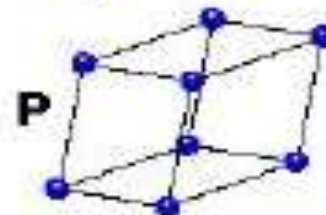
$$\beta \neq 120^\circ$$



TRICLINIC

$$a \neq b \neq c$$

$$\alpha \neq \beta \neq \gamma \neq 90^\circ$$



4 Types of Unit Cell

P = Primitive

I = Body-Centred

F = Face-Centred

C = Side-Centred

+

7 Crystal Classes

→ **14 Bravais Lattices**

Miller Indices

- Miller Indices are a method of describing the orientation of a plane or set of planes within a lattice in relation to the unit cell. They were developed by [William Hallows Miller](#).
- These indices are useful in understanding many phenomena in materials science, such as explaining the shapes of single crystals, the form of some materials' microstructure, the interpretation of X-ray diffraction patterns, and the movement of a dislocation , which may determine the mechanical properties of the material.

Miller Indices

Miller indices are used to specify directions and planes.

- These directions and planes could be in lattices or in crystals.
- The number of indices will match with the dimension of the lattice or the crystal. •

E.g. in 1D there will be 1 index and 2D there will be two indices etc.

Notation Summary

- (h,k,l) represents a point – note the exclusive use of commas
- Negative numbers/directions are denoted with a bar on top of the number
- $[hkl]$ represents a direction
- $\langle hkl \rangle$ represents a family of directions
- (hkl) represents a plane
- $\{hkl\}$ represents a family of planes

Miller Indices for Directions

- A vector $\mathbf{r}$ passing from the origin to a lattice point can be written as:

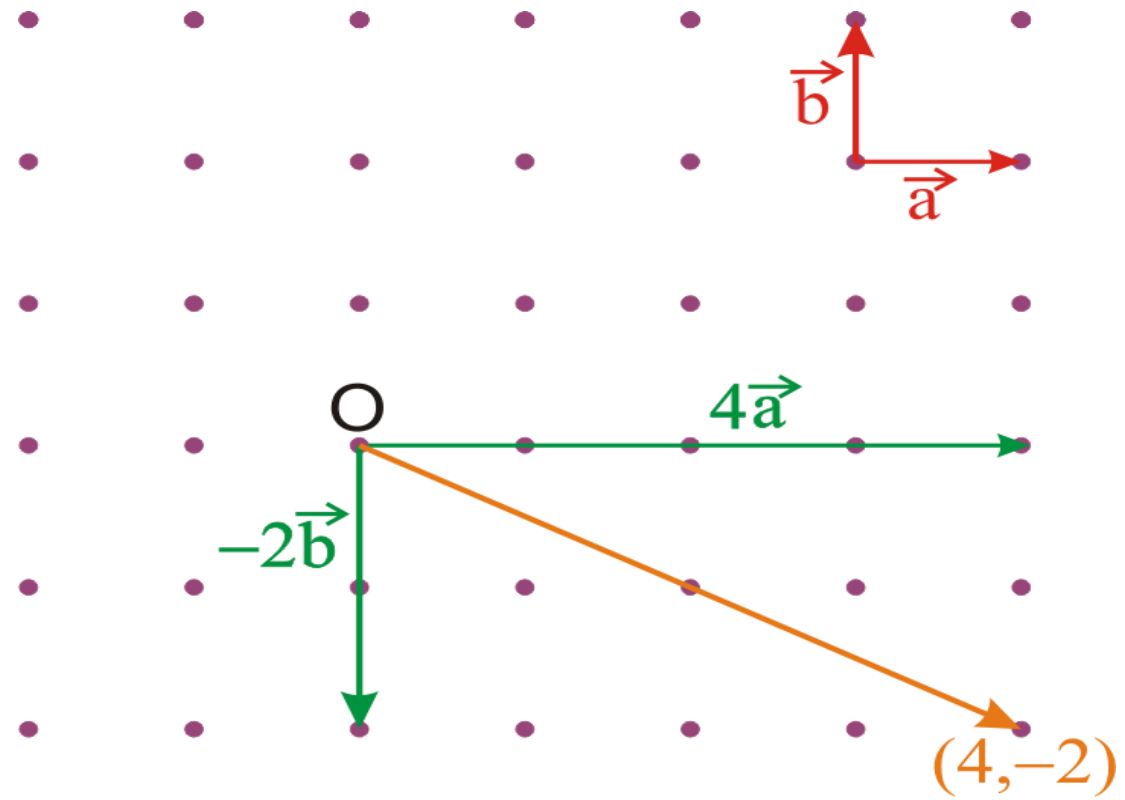
$$\mathbf{r} = r_1 \mathbf{a} + r_2 \mathbf{b} + r_3 \mathbf{c}$$

where, $\mathbf{a}, \mathbf{b}, \mathbf{c} \rightarrow$ basic vectors and

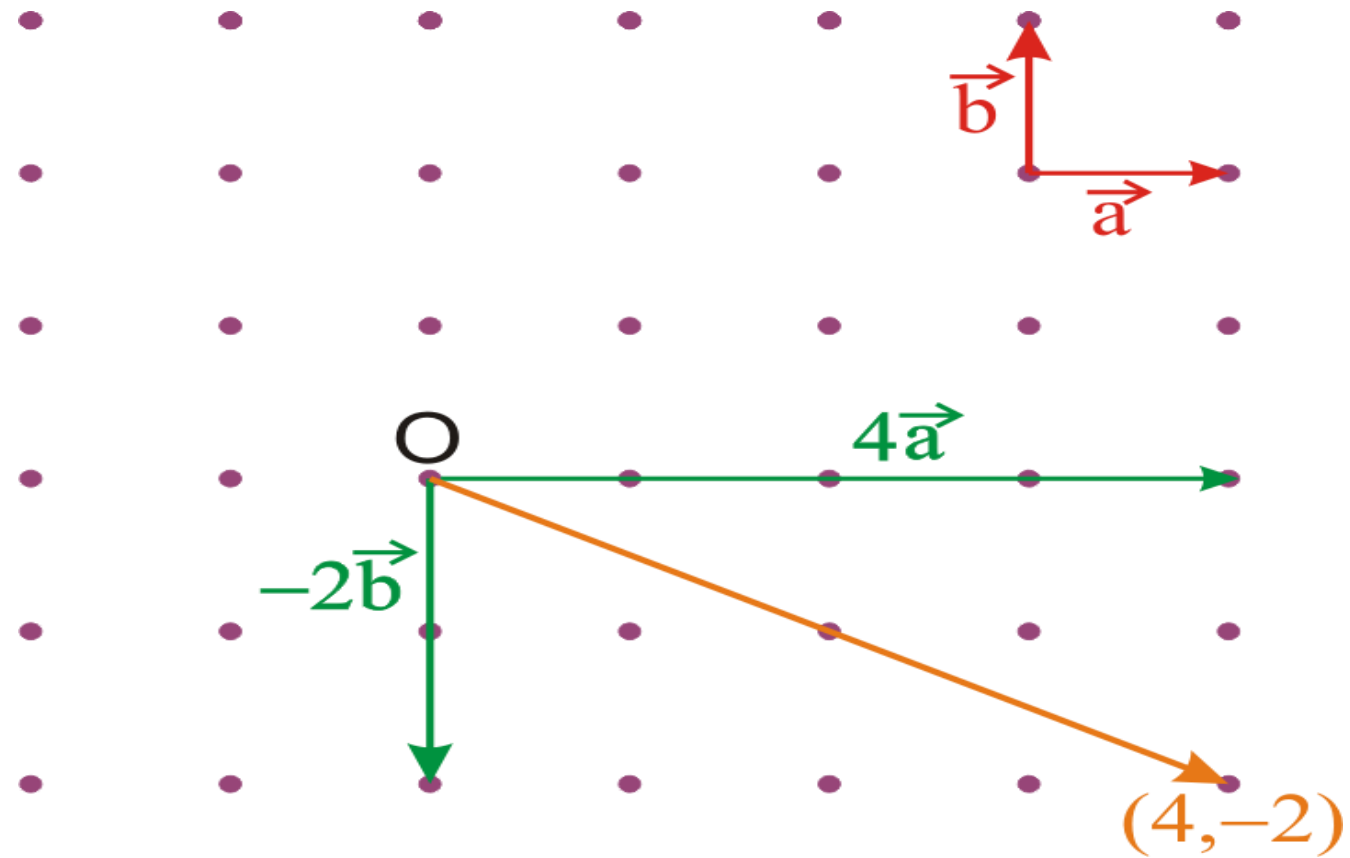
Miller indices $\rightarrow (r_1 r_2 r_3)$

- Fractions in $(r_1 r_2 r_3)$ are eliminated by multiplying all components by their common denominator.

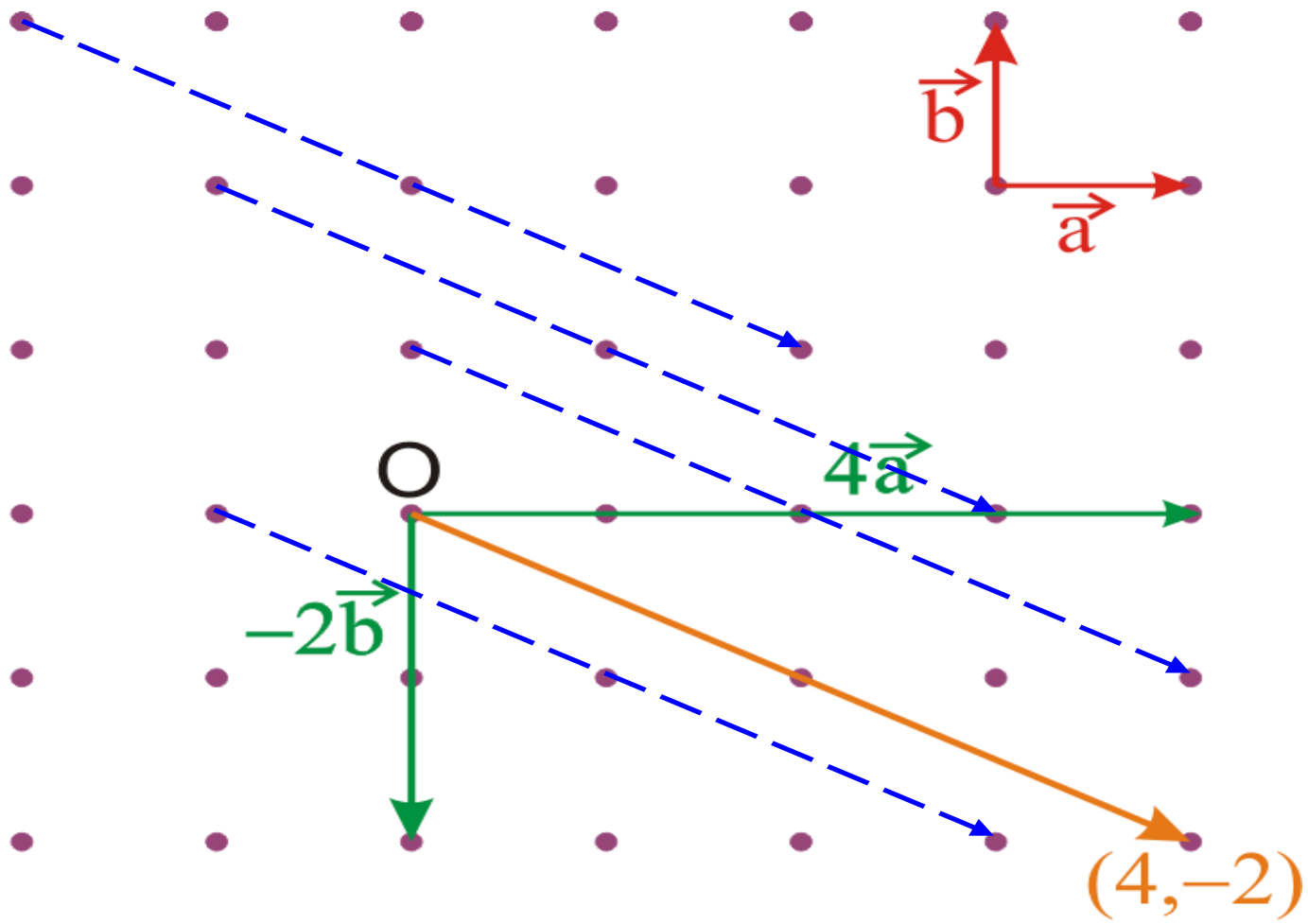
Example



Cont.



Cont.

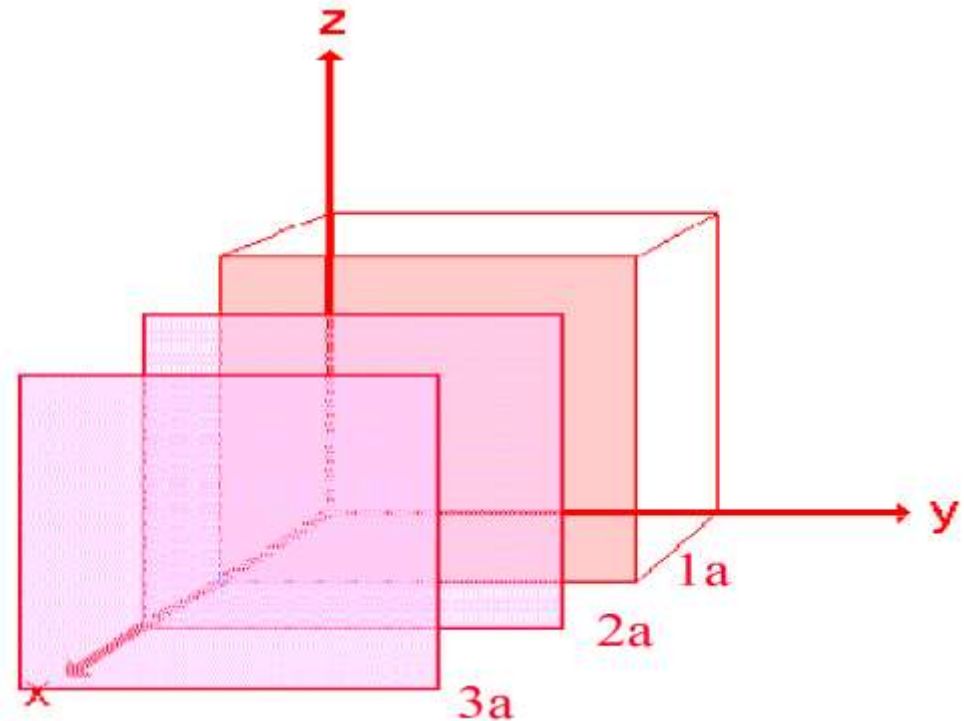


Miller Indices for Planes: Procedure

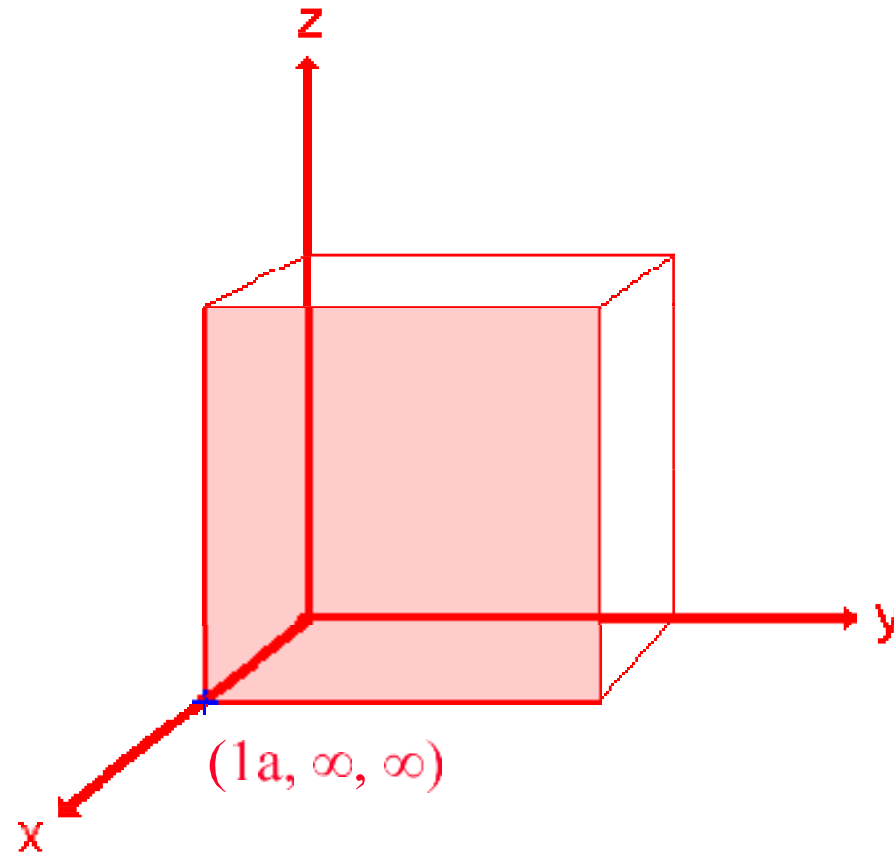
- Identify the plane intercepts on the x, y and z-axes.
- Specify intercepts in fractional coordinates.
- Take the reciprocals of the fractional intercepts.

Miller Indices for Planes: Illustration

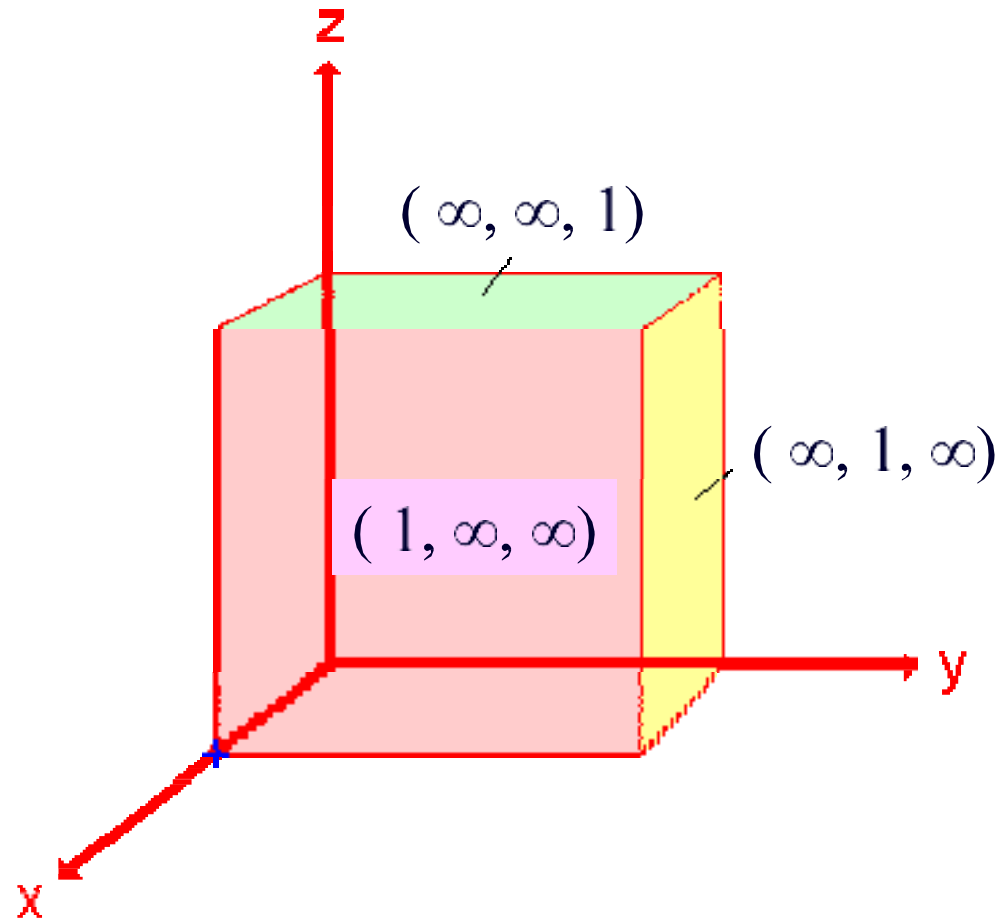
Consider the plane in pink, which is one of an infinite number of parallel planes each a consistent distance (“ a ”) away from the origin (purple planes)



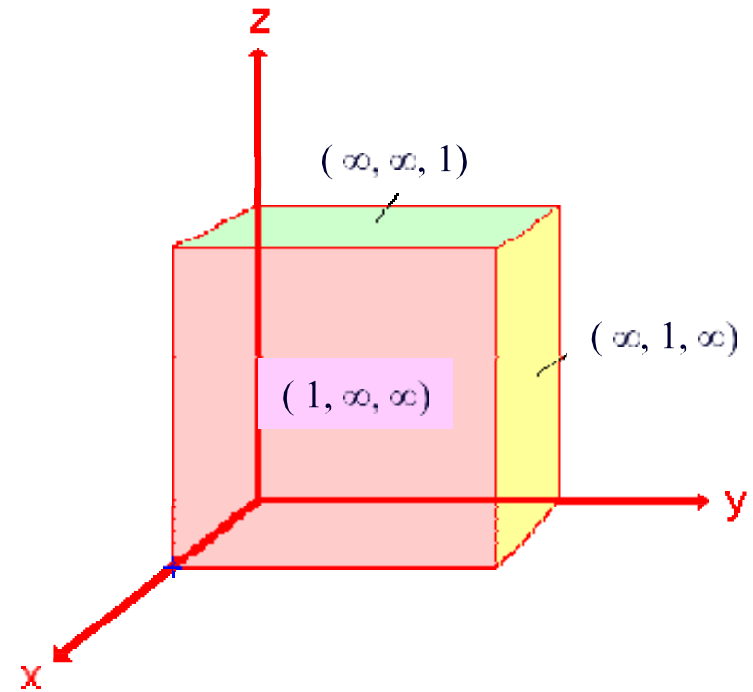
- The plane intersects the x-axis at point a. It runs parallel along y and z axes.
- Thus, this plane can be designated as $(1, \infty, \infty)$



- Likewise, the yellow plane can be designated as $(\infty, 1, \infty)$
- And the green plane can be written as $(\infty, \infty, 1)$



- Miller Indices are the reciprocals of the parameters of each crystal face.
- Pink Face= $(1/1, 1/\infty, 1/\infty) = (100)$
- Green Face= $(1/\infty, 1/\infty, 1/1) = (001)$
- Yellow Face= $(1/\infty, 1/1, 1/\infty) = (010)$



Family of Directions

- It's a set of directions related by symmetry operations of the lattice.

Index	Members in family for cubic lattice
$\langle 100 \rangle$	<i>6 different directions belong to $\langle 100 \rangle$ family:</i> $[100], [\bar{1}00], [010], [0\bar{1}0], [001], [00\bar{1}]$
$\langle 110 \rangle$	<i>12 different directions belong to $\langle 110 \rangle$ family:</i> $[110], [\bar{1}10], [1\bar{1}0], [\bar{1}\bar{1}0], [101], [\bar{1}01],$ $[10\bar{1}], [\bar{1}0\bar{1}], [011], [0\bar{1}1], [01\bar{1}], [0\bar{1}\bar{1}]$
$\langle 111 \rangle$	<i>8 different directions belong to $\langle 111 \rangle$ family:</i> $[111], [\bar{1}\bar{1}\bar{1}], [1\bar{1}\bar{1}], [\bar{1}1\bar{1}], [\bar{1}\bar{1}1], [1\bar{1}1], [11\bar{1}], [\bar{1}11]$

Crystal Imperfection

- A perfect crystal, with every atom of the same type in the correct position, does not exist.
- There always exist crystalline **defects**, which can be **point defects** occurring at a single lattice point; **line defects** occurring along a row of atoms; or **planar defects** occurring over
- A two-dimensional surface in the crystal. There can also be three-dimensional defects such as **voids**.

Crystalline Defects:

Imperfections or defects:

- Any deviation from the perfect atomic arrangement in a crystal is said to contain imperfections or defects. Or a crystalline defect is a lattice irregularity having one or more of its dimensions on the order of an atomic dimension.

There are 4 major categories of crystalline defects:

- **Zero dimensional: Point defects** occurring at a single lattice point.
- **One dimensional: Linear defects (dislocations)** occurring along a row of atoms.
- **Two dimensional: Planar (surface) defects** occurring over a two-dimensional surface in the crystal.
- **Three dimensional: Volume (bulk) (void) defects**

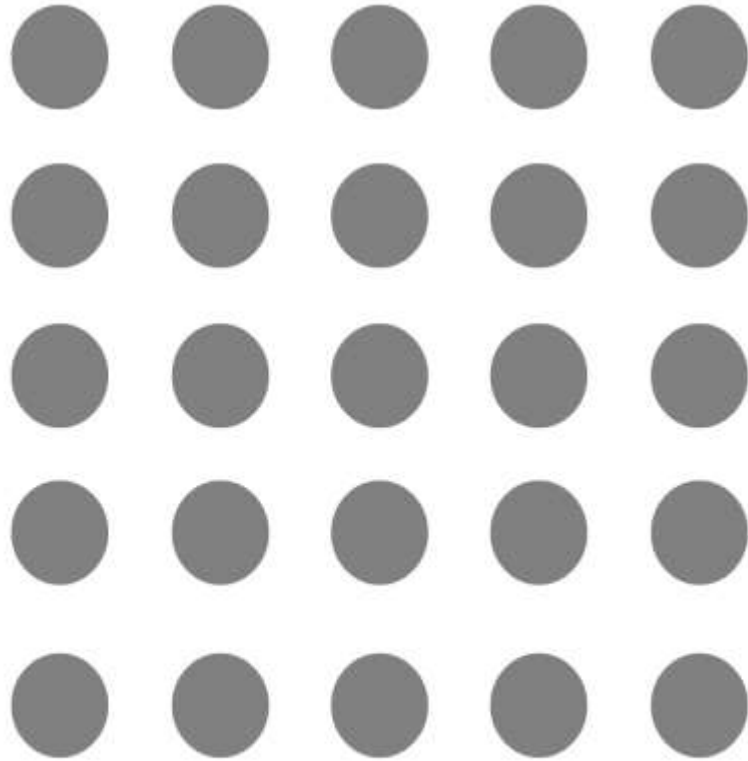
Crystal Defects Classification:

Point defects

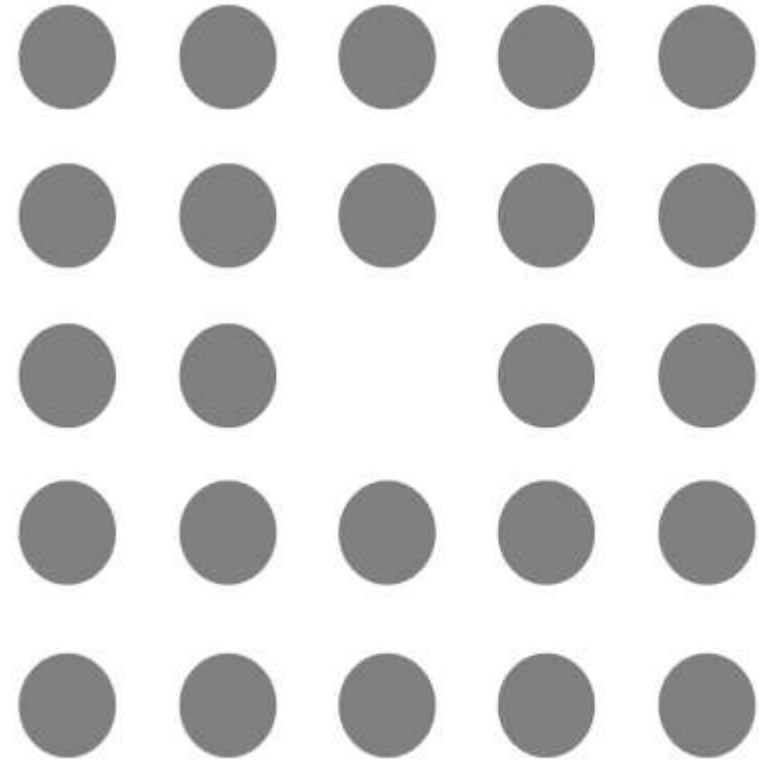
- Vacancy
- Schottky
- Self interstitial
- Frenkel
- Colour centers
- Polarons
- Excitons

Point Defects: Point defects are where an atom is missing or is in an irregular place in the lattice structure

Vacancies: A perfect crystal with regular arrangement of atoms can not exist. There are always defects, and the most common defects are point defects. This is especially true at high temperatures when atoms are frequently and randomly change their positions leaving behind empty lattice sites, called **vacancies**. Or **Vacancies** are empty spaces where an atom should be, but is missing. In most cases diffusion (mass transport by atomic motion) - can only occur because of vacancies.



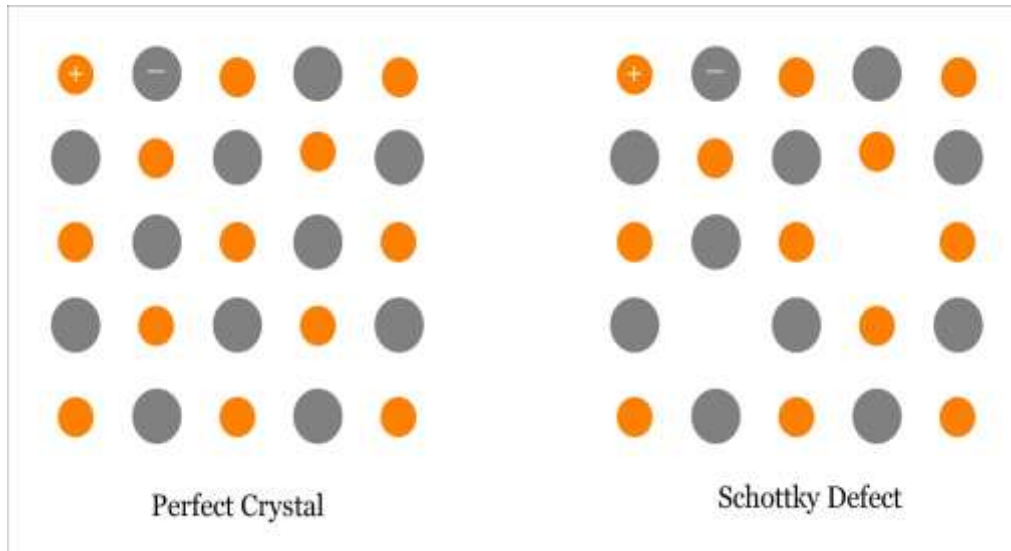
Perfect Crystal



Vacancy - Point Defect

Schottky defect:

It is a type of vacancy in which an atom being free from regular site, migrates through successive steps and eventually settles at the crystal surface. a pair of anion and cation vacancies.

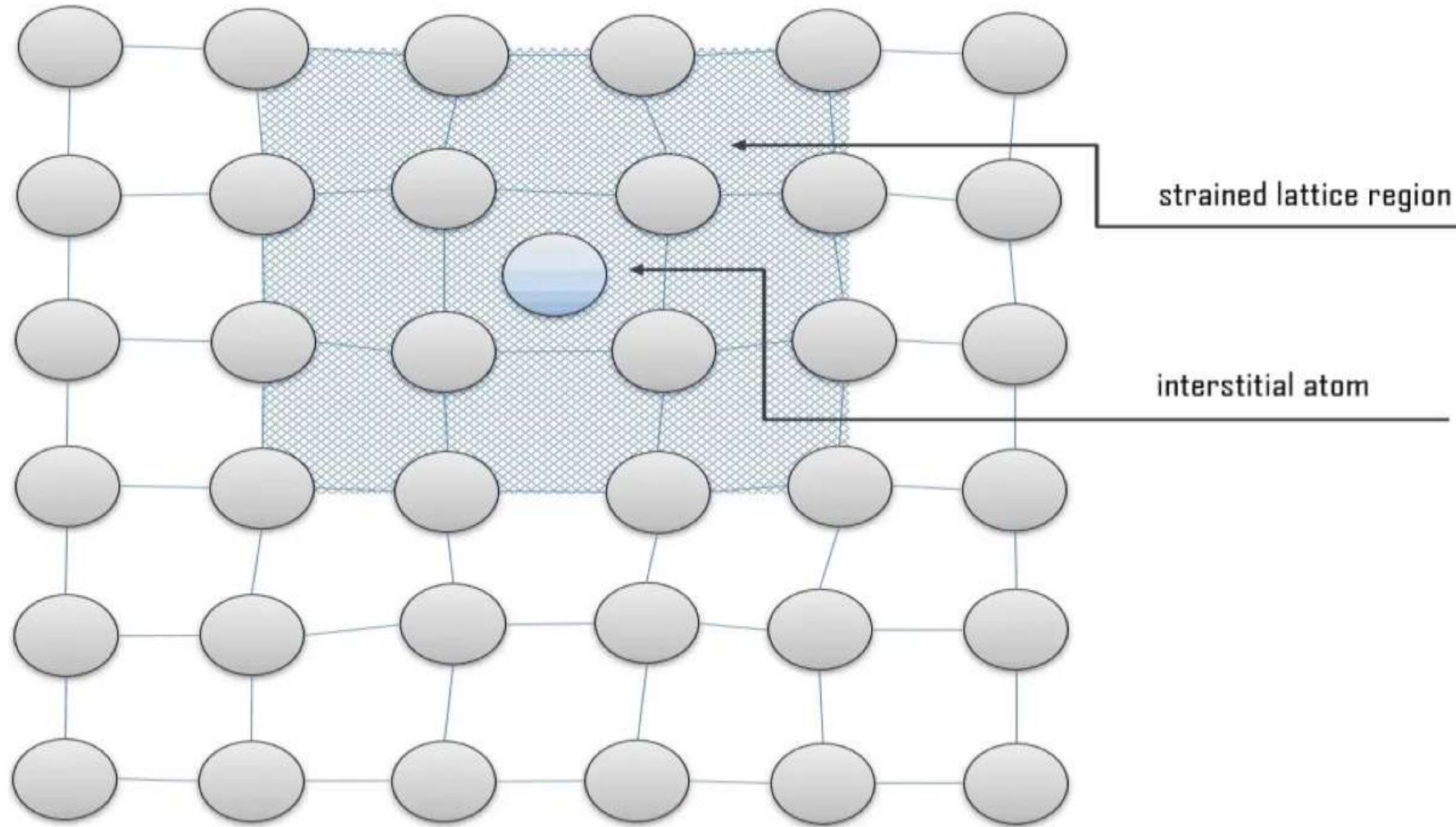


Interstitials:

Atoms that are squeezed in between regular lattice sites. If the interstitial atom is of the same species as the lattice atoms, it is called **self-interstitial**. Creation of a self-interstitial causes a substantial distortions in the surrounding lattice and costs more energy as compared to the energy for creation of a vacancy ($Q_i > Q_v$) and, under equilibrium conditions, self-interstitials are present in lower concentrations than vacancies.

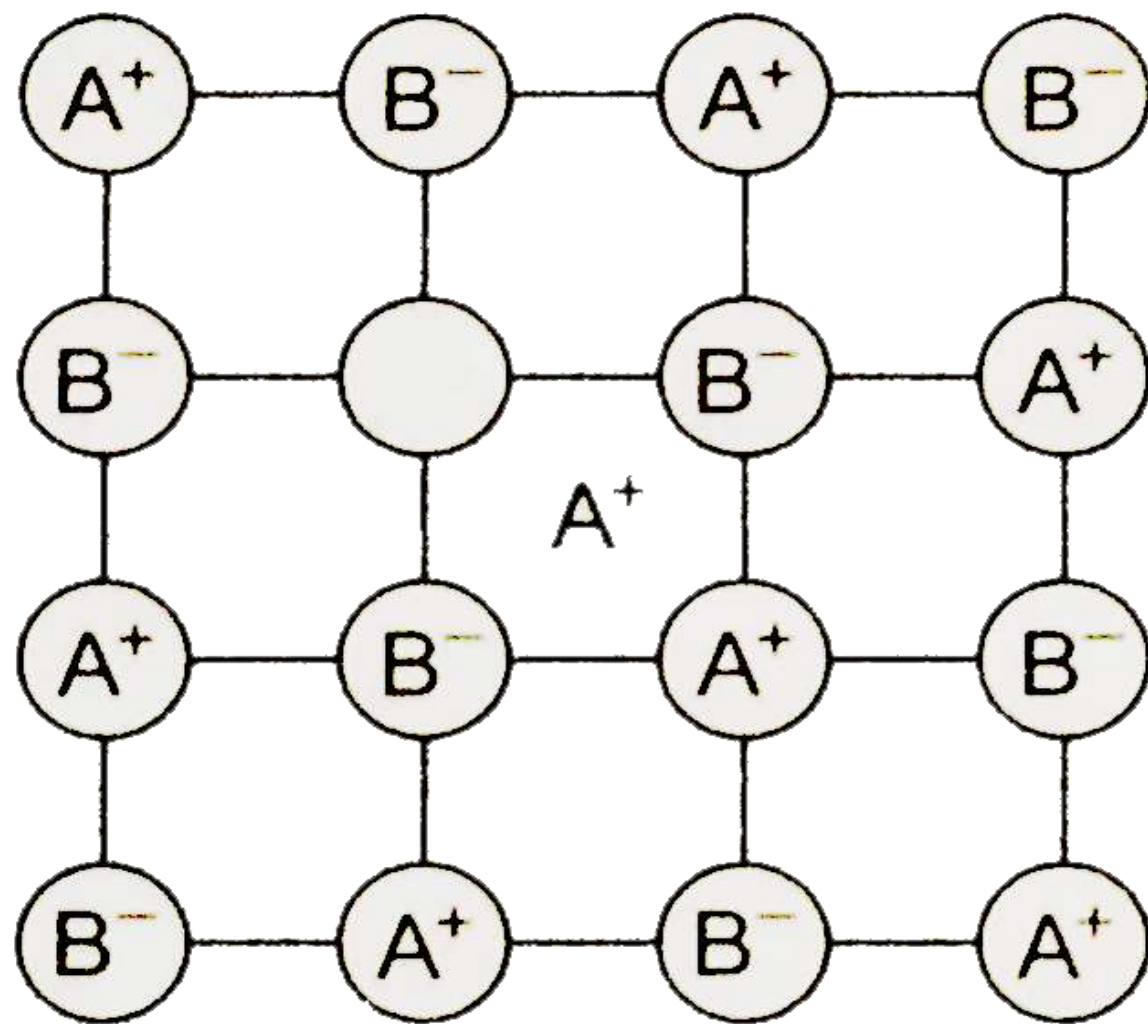
- Usually smaller atoms (carbon, nitrogen, hydrogen, oxygen) are called **interstitial impurities**. **Interstitial** impurity atoms are much smaller than the atoms in the bulk matrix.
- Interstitial impurity atoms fit into the open space between the bulk atoms of the lattice structure. An example of interstitial impurity atoms is the carbon atoms that are added to iron to make steel.

Interstitial Defect



Frenkel defect:

- A Frenkel defect is a type of point defect in crystalline solids that occurs when an atom or ion leaves its lattice site and occupies a nearby interstitial site, creating a vacancy. It's also known as a dislocation defect or a Frenkel pair or disorder.
- Frenkel defects are common in ionic compounds with large differences in the size of the cation and anion, and low coordination numbers. Some examples of compounds that exhibit Frenkel defects include silver bromide, silver chloride, silver iodide, and zinc sulfide.



Cation : A^+

Anion : B^-

Line Defect:

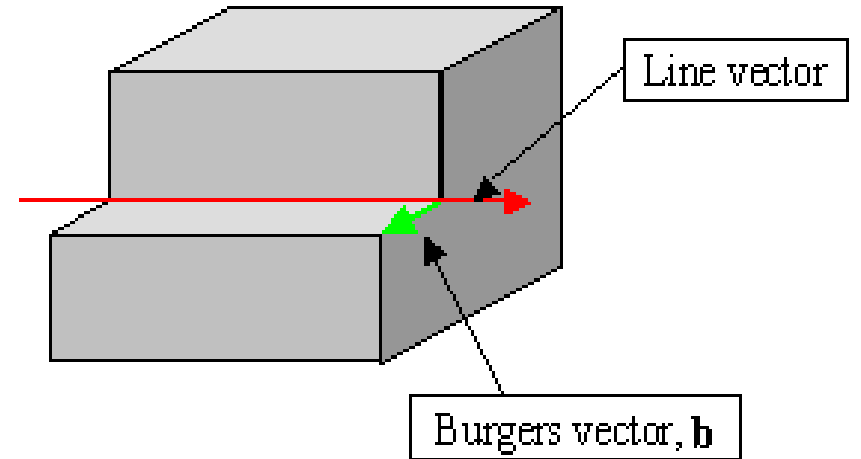
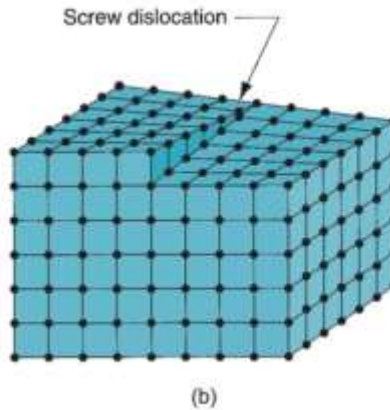
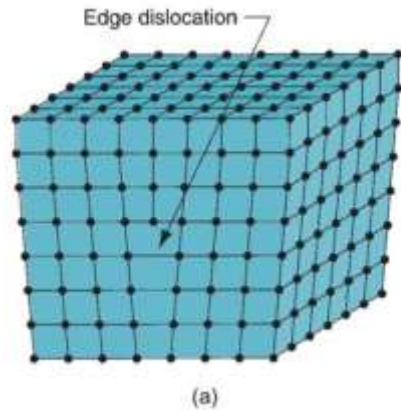
- A line defect in a crystal, also known as a dislocation, is a one-dimensional irregularity in the crystal lattice that occurs when a row or line of lattice points is misaligned or distorted.

Line defects can be classified as

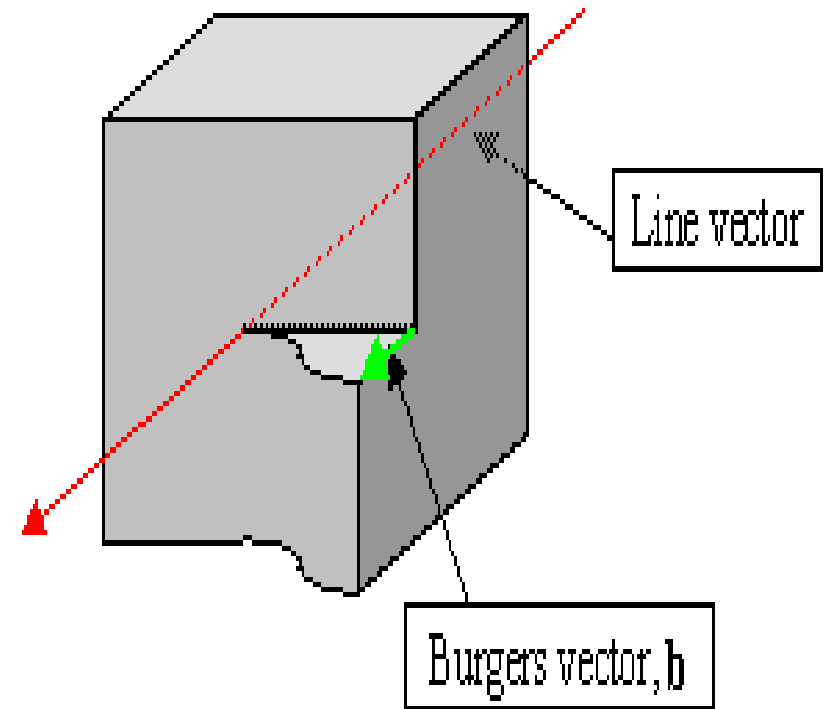
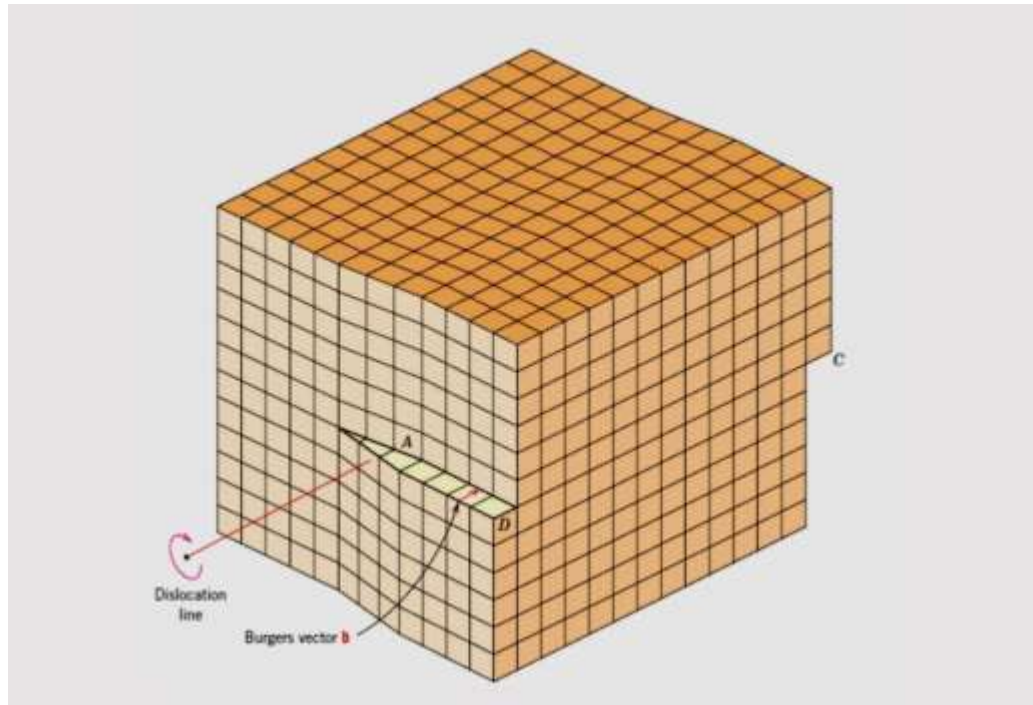
1. Edge dislocations
2. Screw dislocations

They can affect the mechanical behavior, deformation, and plasticity of materials, and can potentially weaken or strengthen a solid.

Edge Dislocation: An edge dislocation occurs when an extra plane or layer of atoms extends part way into the crystal, which causes atoms in that region of the crystal to be compressed, but in the region where the extra plane does not extend, they are spread apart.

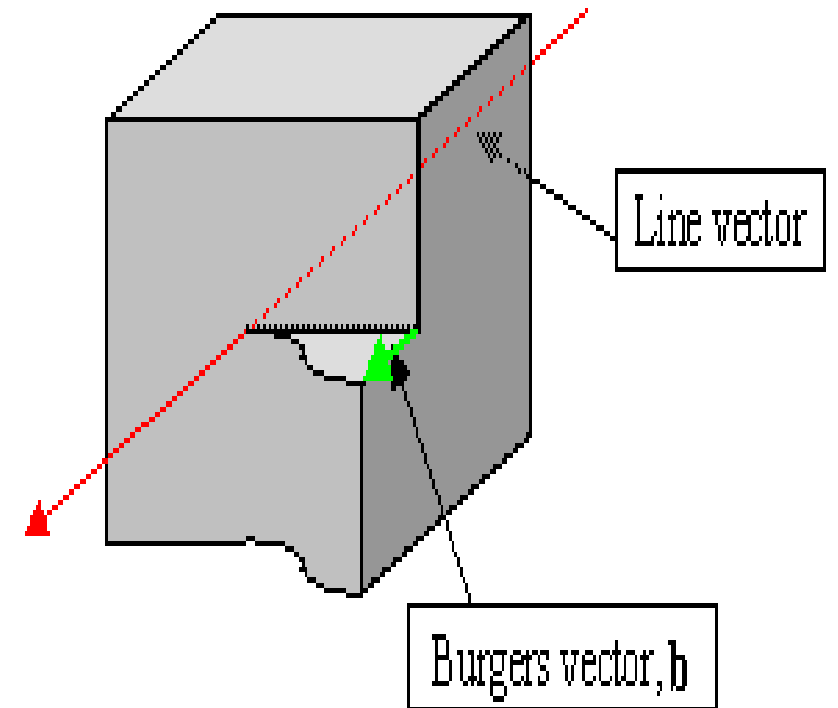
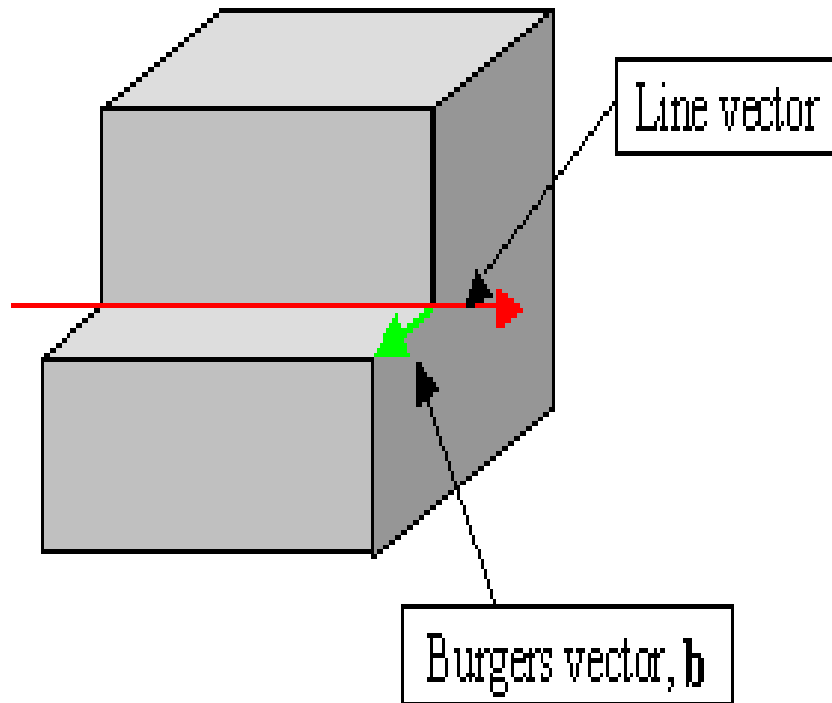


Screw dislocations: A screw dislocation is a type of crystal defect that occurs when one half of a crystal is slid across the other, resulting in a spiral stacking of crystal planes around the center line



Burgers vector : In [materials science](#), the **Burgers vector**, named after Dutch physicist [Jan Burgers](#), is a [vector](#), often denoted as **b**, that represents the [magnitude](#) and direction of the lattice distortion resulting from a [dislocation](#) in a [crystal lattice](#)

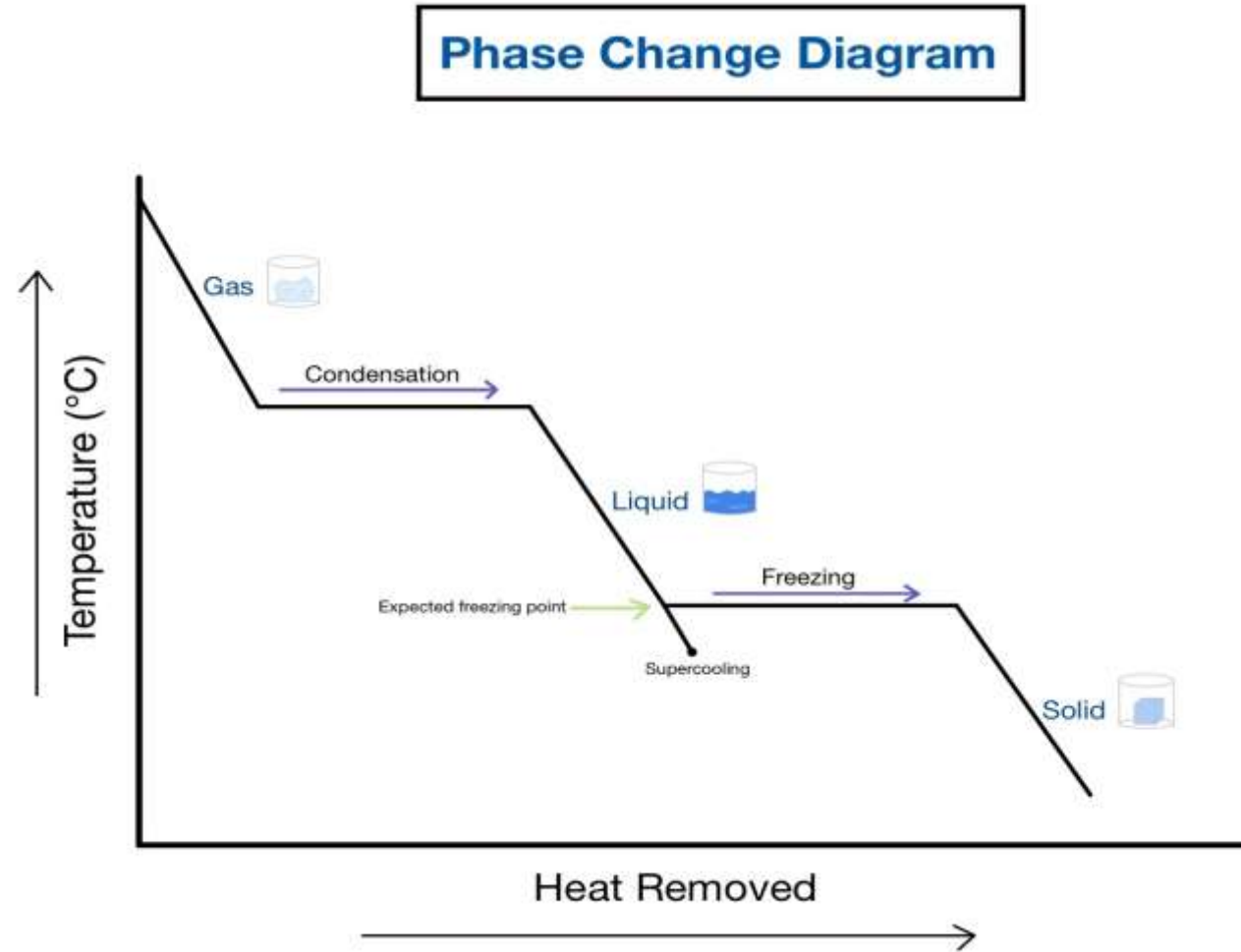
In edge dislocations, the Burgers vector and dislocation line are perpendicular to one another. In screw dislocations, they are parallel

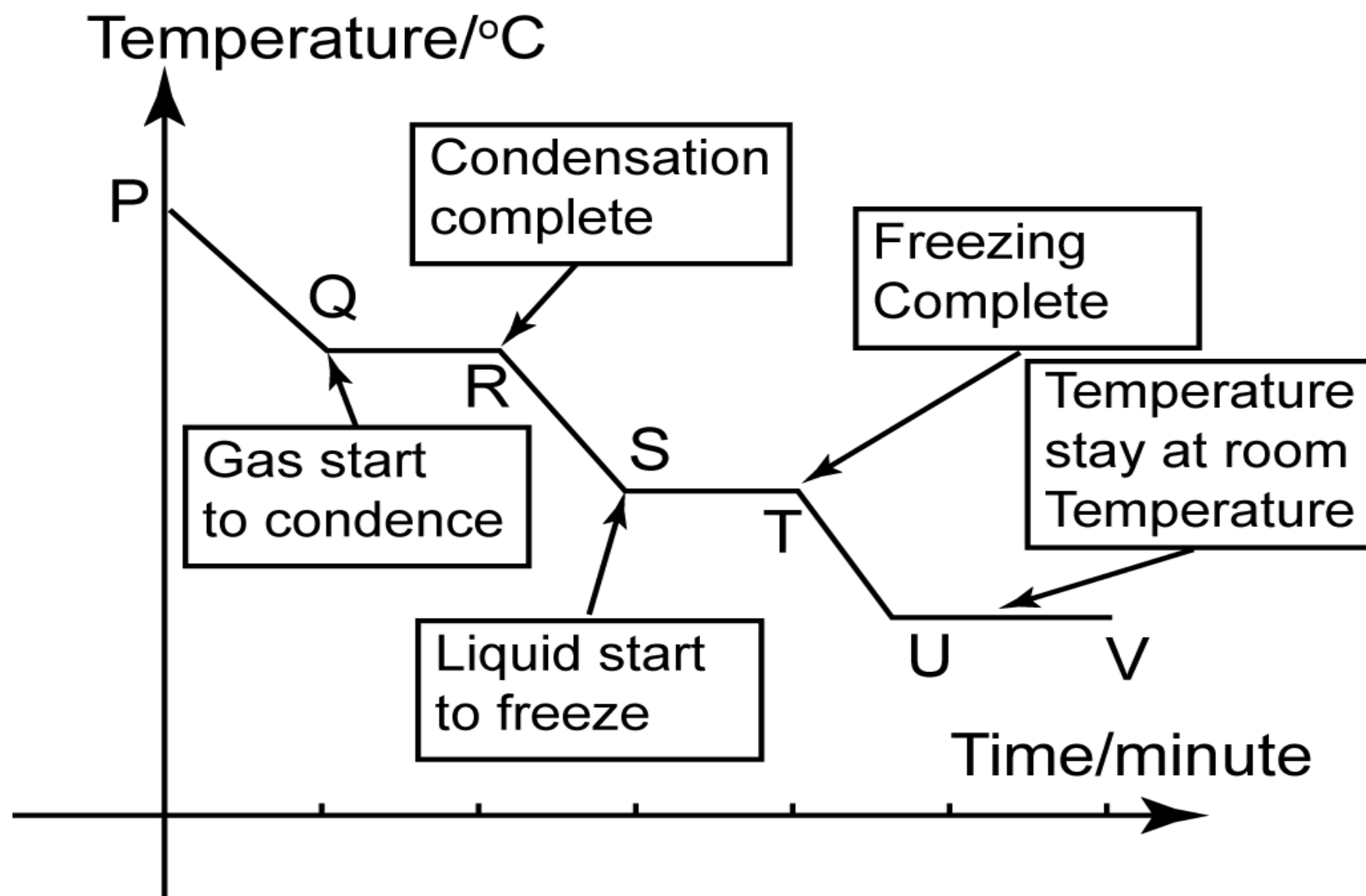


Cooling curve

- The cooling curve of a substance is a graph that shows how its temperature changes over time as it cools. The gradient of the cooling curve is related to the substance's heat capacity, thermal conductivity, and the external temperature.
- The shape of a cooling curve can be continuous or discontinuous. It's continuous if the material cools slowly from a liquid state.

COOLING CURVE





Nucleation

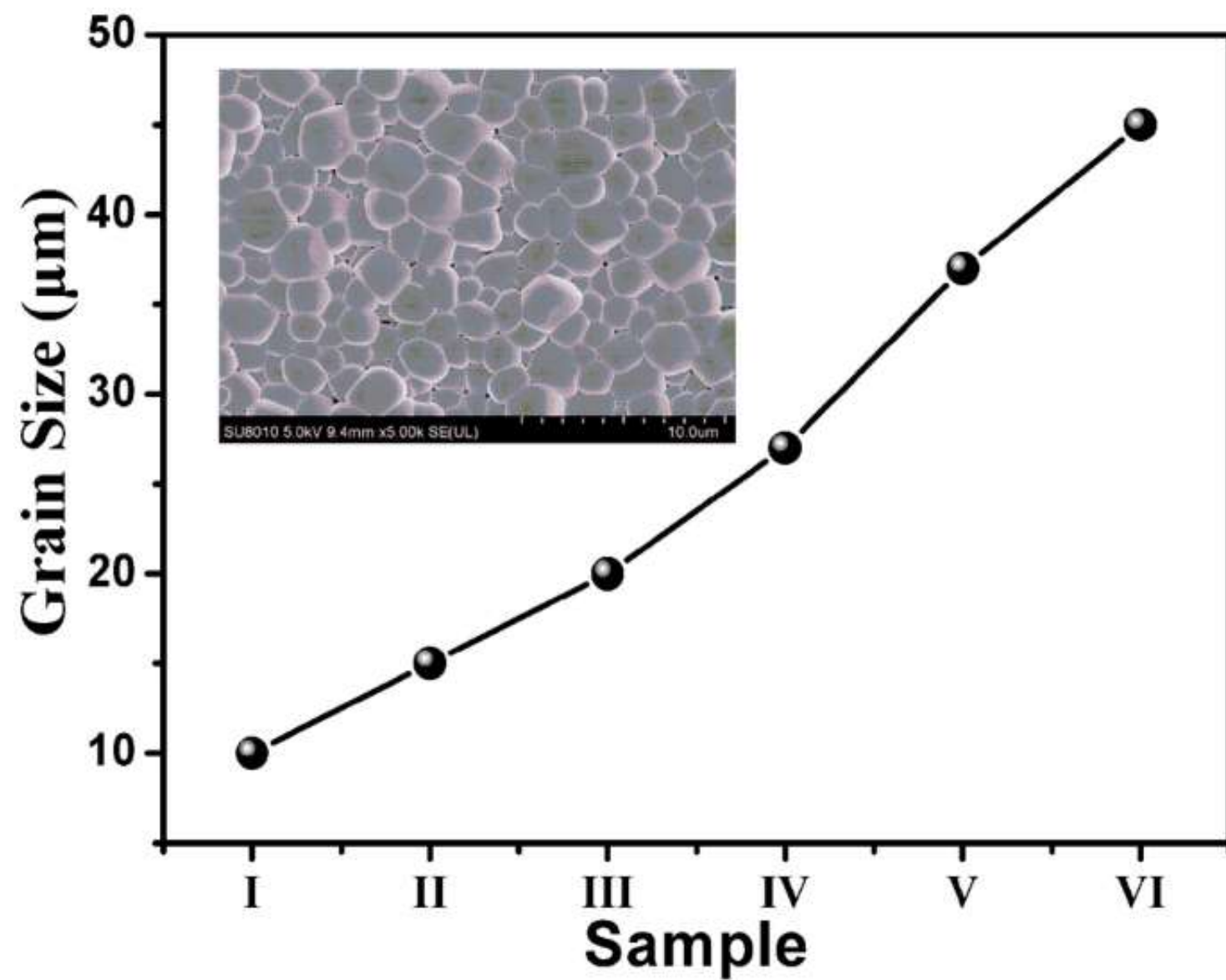
- Nucleation of crystals is the initial step in the process of crystal formation, where a small number of atoms, molecules, or ions arrange themselves into a crystalline pattern. This process can occur in a liquid, solution, or vapor. The resulting structure, called a nucleus or embryo, can then grow into a macroscopic crystal.

•



Grain growth

- Grain growth is the increase in size of grains in a material at high temperatures. It's a common phenomenon in polycrystalline materials, such as metals and ceramics, that occurs during thermal treatments like sintering.
- Grain growth is important because it controls the kinetics of many phase transformations and recrystallization processes. It's also used in materials science and engineering to understand the behaviors of grains, which is analogous to coarsening.



THANKS